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Infrasync : Blockchain and Machine Learning Based System for Transparent Fund Allocation and Road Construction Monitoring

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CERTIFICATE

This is to certify that the project report entitled

INFRASYNC: BLOCKCHAIN AND MACHINE LEARNING BASED SYSTEM FOR TRANSPARENT FUND ALLOCATION AND ROAD CONSTRUCTION MONITORING

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ABSTRACT

There are quite a number of obstacles that impede road construction processes. Overspending of budgets, poor resource management, inefficiency in allocating money and tracing expenses – all this occurs too frequently. In my opinion, it is a major drawback since it results in waste of funds and probably corruption. There are attempts to apply machine learning to forecasting costs and blockchain to making financial transactions. However, such techniques are rarely implemented together, while this concept suggests integrating both into one solution. Thus, combining predictive analytics and transparency due to blockchain should contribute to efficiency and accountability to some extent. What I like about the paper is its approach towards data gathering. This dataset is designed from scratch, containing 1000 records and 27 features. Sources are various public resources such as comprehensive report of a project and survey results. The lack of standardized data collection techniques creates a necessity for this dataset creation. It allows training models properly without having to search elsewhere. As far as forecasting is concerned, the technique that is applied is called Extreme Gradient Boosting. It is capable of handling complex non-linear connections between project characteristics, producing accurate cost predictions based on the real project. Again, although it might be questionable whether it works perfectly, it is reliable, in any case. Blockchain-based transactions record allocation of funds, winners of procurement contracts, and transactional flows in general. All processes are secured by means of smart contracts to trace any activity and ensure transparency that contributes to securing record management procedures. Results obtained during testing were remarkable, with high prediction accuracy. Implementation of this method in an integrated model increases trust in development of infrastructures. In total, this framework looks promising in the context of building transparent data-driven and corruption-proof road construction projects. Yet, implementing it everywhere may require further efforts.

CONTENTS

1 Introduction **1**

1.1 Overview 1

1.2 Motivation 2

1.3 Problem Statement and Objectives 3

1.3.1 Problem statement 3

1.3.2 Objectives 3

1.4 Scope 4

1.5 Methodologies of Problem Solving 5

2 Literature Survey **7**

2.1 Review of Recent Literature 7

2.2 Gap Identification/ Common Findings From Literature 9

3 Software Requirements Specifications **10**

3.1 Functional Requirements 10

3.1.1 System Feature 1: Blockchain-based Fund Allocation 10

3.1.2 System Feature 2: Milestone-based Payment System 10

3.1.3 System Feature 3: Predicting Construction Cost using Machine Learning 11

3.1.4 System Feature 4: User Management System 11

3.1.5 System Feature 5: Project Management System 12

3.2 External Interface Requirements 12

3.2.1 User Interfaces 12

3.2.2 Hardware Interfaces 13

3.2.3 Software Interfaces 13

3.2.4 Communication Interfaces 14

3.3 Non-Functional Requirements 14

3.3.1 Performance Requirements 14

3.3.2 Safety and Security Requirements 15

3.4 System Requirements 15

3.4.1 Database Requirements 15

3.4.2 Software Requirements (Platform Choice) 15

3.4.3 Hardware Requirements 15

3.5 SDLC Model to Adopt 16

4 System Design **17**
4.1 System Architecture . . . 17
4.2 **Data Set / Database Structure** . . . 20
4.2.1 Database Design (MongoDB) . . . 20
4.2.2 Blockchain Storage . . . 21
4.2.3 Dataset for ML model: . . . 22
4.3 Mathematical Model . . . 22
4.3.1 Input and Output Representation . . . 22
4.3.2 Prediction Function: . . . 23
4.3.3 Objective Function . . . 23
4.3.4 Model Optimization . . . 24
4.3.5 Importance of the Model . . . 24
4.4 UML Diagrams . . . 25
4.4.1 Use case diagram . . . 25
4.4.2 Class Diagram . . . 26
4.4.3 Sequence Diagram . . . 27
4.4.4 Activity Diagram . . . 28

5 PROJECT PLAN **30**
5.1 Risk Management . . . 30
5.1.1 Risks Identification . . . 30
5.1.2 Risks Analysis . . . 30
5.1.3 Risks Mitigation, Monitoring, and Management . . . 30
5.2 Project Schedule . . . 31
5.2.1 Set of Project Tasks . . . 31
5.2.2 Timeline Graph . . . 32
5.3 Team Structure . . . 32

6 Project Implementation **34**
6.1 Overview of Project Modules . . . 34
6.2 Tools and Technologies Used . . . 36
6.3 Detailed Description of Algorithms . . . 36
6.3.1 Algorithm 1: Blockchain-based Fund Allocation and Payment . . . 37
6.3.2 Algorithm 2: Predict Estimated Cost using Machine Learning . . . 37
6.3.3 Algorithm 3: Authenticate User and Manage Roles . . . 38
6.3.4 Algorithm 4: Project Creation and Monitoring . . . 38

7 Software Testing **40**
7.1 Types of Testing . . . 40

7.2 Test Cases and Test Results 42

8 Results and Discussions **44**
8.1 Findings 44
8.2 Results Analysis and Validations 46
8.3 Screenshots 50

9 CONTRIBUTION TO SUSTAINABLE DEVELOPMENT GOALS (SDGs) **55**
9.1 Introduction to SDGs 55
9.2 Mapping of the Project to Relevant SDGs 55
9.3 Evaluation of the Impact of the Project 56
9.4 Issues and Future Scope 57

10 Conclusion & Future Scope **58**
10.1 Conclusion 58
10.2 Future Work 58
10.3 Applications 60

Appendix A: Details of The Research Paper **63**
A.1 Paper Details 63
A.2 Proof of Submission 63

Appendix B: Plagiarism Report **65**

References **70**

LIST OF FIGURES

4.1 System Architecture of the Proposed Model 19
4.2 Use Case Diagram 25
4.3 Class Diagram 26
4.4 Sequence Diagram 27
4.5 Activity Diagram 28

5.1 Timeline Graph 32

8.1 Accuracy graph 47
8.2 Residual Plot 48
8.3 Important Features 49
8.4 Homepage 51
8.5 Tender Creation 51
8.6 Bid Evaluation Interface 52
8.7 Contract Award 52
8.8 Public Transparency Portal 53
8.9 NFT Certificate 53
8.10 Blockchain Transaction 54

A.1 Email Confirmation of Paper Submission 64

LIST OF TABLES

2.1 Literature Survey of Existing Approaches 8

7.1 Test Cases for System Validation 42

Chapter 1

INTRODUCTION

1.1 Overview

The role played by road infrastructure is crucial for the development of any country. An excellent road network will help enhance transportation, reduce traveling time, and facilitate connections within both urban and rural areas. In addition, the road will promote development through industries, commerce, and access to basic needs such as health and education facilities. Thus, the construction of roads can play an important part in the development of a country. In many underdeveloped areas, there are various significant problems regarding the road construction process. The first major challenge is that of budget misallocation and mismanagement of funds allocated for the process because funds may be diverted for personal gains. In addition, there is inefficiency in the allocation process because the implementation of projects is delayed due to improper planning and tracking processes. Finally, there may be low-quality control that results in the need for frequent repair and high maintenance costs. Existing systems for road construction project management are characterized by heavy dependency on manual labor and centralization. They lack any automation capabilities and are prone to various mistakes that can be made by a human, manipulation, and inefficiencies. Due to the fact that information is stored in isolated or opaque systems, the parties involved cannot obtain adequate visibility on the allocation of funds for road construction. Thus, existing systems suffer from low levels of accountability and trust. The second important limitation of current road construction systems is that there are no means for monitoring the progress of the project and its finances. It is impossible to verify the completion of project milestones or the release of funds. Such an approach leads to inadequate governance and ineffective decision-making. In order to address the aforementioned issues, InfraSync: Intelligent & Transparent Road Construction System will be developed. This system will use advanced technologies, such as blockchain and machine learning, to facilitate modern management of infrastructure. The blockchain part is essential for ensuring efficient and transparent management of funds allocated to the projects. The use of smart contracts guarantees automatic and reliable milestone-based payment release which occurs only after the completion of milestones by the contractor. The transactions conducted using this technology are permanently stored in the immutable ledger which eliminates any chances of manipulating the data. Also, the platform uses various machine learning algorithms in order to estimate

the cost of the construction project. As the result, the cost prediction becomes rather accurate taking into account all the necessary input parameters like the price of materials, labor needs, project scale etc. Authorities will receive useful insights from the model and, thus, be able to make better decisions. Moreover, InfraSync was designed in the form of a web-based platform which allows interacting with the system effectively. The system offers various functionalities like the project creation, funds distribution, management of milestones, and transaction monitoring for different users – governmental authorities and contractors. Summing up, InfraSync offers a complete and highly advanced solution to cope with the issues related to road construction management.

1.2 Motivation

The main rationale that forms the basis for InfraSync system stems from the persistent problems experienced in infrastructure projects, especially in the developing world. Although huge sums of money are being invested by governments on infrastructure development projects, the desired goals are not always attained due to inefficiencies in project management and implementation processes. The inadequacy in the current approaches calls for innovative ways to manage and implement such projects. One of the major problem areas in the current projects is the misappropriation of the funds allocated towards the development work. This is because there is no proper way of ensuring that the funds have been properly accounted for during the course of their use. It is only through lack of transparency in fund management and distribution that corrupt practices become rampant in project management. Another equally important problem is the absence of effective tools for monitoring and controlling. Conventional solutions have much in common with manual accounting and control, which can lead to the distortion of information about the current situation within the project. In consequence, potential problems and inefficiencies are detected too late and thus cannot be addressed in time, causing an increase in expenses and time spent on work. Finally, the inadequacy of advanced methods for planning and estimation of cost factors should be mentioned. The process of cost estimation is commonly associated with traditional manual calculations. Such approaches tend to produce incorrect estimations due to the lack of reliable information. This leads to the overestimation of budgetary resources, poor planning, and further inefficiencies. However, due to the development of the new generation of technologies, it becomes feasible to solve these issues. For example, using blockchain can be an effective solution, since it provides an opportunity to manage all financial transactions in a decentralized way without any possibility of any fraud due to the immutability and transparency of transactions in this case. Furthermore, using smart contracts is the best decision in this situation because it allows

automation of payment processes in the course of fulfillment of predefined conditions. At the same time, machine learning methods offer opportunities for processing data and forecasting some situations in the future, thus allowing creating a more reliable model for estimating the cost. It would become a good decision to implement such models in order to minimize risks associated with the estimation process. The motivation for conducting this research is connected with combining the two types of technologies in the course of designing a new system.

1.3 Problem Statement and Objectives

1.3.1 Problem statement

Presently, there are some significant problems that hamper the management of road construction projects in many areas. These problems are mostly associated with the lack of transparency, poor utilization of finances, and improper management processes. The current systems involve manual methods of work and a centralized approach that often cause low visibility among the participants. First of all, the management of road construction projects lacks an efficient way of managing money spent on their realization. Finances are often used without any control, which can contribute to misuse of funds and fraud. As a consequence, many projects end up with financial losses and unfinished tasks. Secondly, besides poor use of finances, a lack of efficient tools for monitoring and controlling the process becomes an additional challenge for the effective management. Authorities cannot easily track the progress and determine whether certain goals were achieved timely. The other issue that should be mentioned is lack of reliable cost estimation techniques, which would provide reliable figures. Traditionally, cost estimates are not very accurate due to various assumptions made during their calculation process. As a result, there are costs overruns and misallocation of funds and resources. Thus, it is needed to develop a system that will guarantee financial transaction reliability, will help track the work progress effectively, and allow for proper cost prediction.

1.3.2 Objectives

- **To design and develop a transparent system of financing using blockchain technologies**: This objective implies developing a transparent funding mechanism by recording transactions into a decentralized ledger where any attempt at data manipulation will be impossible.
- **To design and use smart contracts for funding at each project milestone stage**

: This means creating mechanisms of automatic funding upon completion and verification of each milestone stage in the process of construction.

- **To apply machine learning algorithms (XGBoost) for predicting construction cost:** This implies using appropriate machine learning algorithms for analysis of relevant parameters and prediction of construction costs.
- **To increase efficiency and transparency of road construction projects**: In other words, this objective is associated with achieving higher efficiency, avoiding corrupt practices, and minimizing delay periods.
- **To create a central platform for project management:** : This implies building a web-based application that will allow users to manage projects and monitor their progress.

1.4 Scope

The project scope of the InfraSync project involves the development of a prototype of an integrated system involving technologies including blockchain and machine learning. It is aimed at overcoming the problems involved in managing road construction projects such as inefficiency of fund management, absence of transparency, and inaccurate cost estimation through the introduction of innovative technology solutions. The initial scope of the project entails creating a blockchain-based system for fund management in which all the transactions concerning the construction projects will be entered into a tamper-proof ledger in order to enable effective tracking of fund allocation and usage. The second key component of the system is the use of smart contracts in order to automate the transfer of funds. The payment of money depends on the project milestones which are set, and only after successfully reaching them are the payments transferred to the recipients.Besides the financial transparency aspect, the system also features a machine learning algorithm to predict the cost of the construction. The algorithm evaluates several parameters such as costs of materials, labor needs, and scale of the construction project to provide an estimate of the cost of construction. The integration of prediction algorithms will improve the process of planning and budgeting, lowering the possibility of cost overrun issues. The system is a web-based application that allows individuals such as the government and construction contractors to interact with the system. The system features several functions such as the ability to create projects, track milestones, allocate funds, and monitor transactions. Nevertheless, the project scope only includes a prototype model of the system. The system showcases how blockchain and machine learning can be used in government-funded construction but does not feature a large-scale application of the

technology. Furthermore, there is no full-scale integration with the existing databases, legal regulations, and frameworks within the government domain. Also, considerations such as policy limitations, massive scaling, and integration of field data into real time cannot be achieved at the moment by this implementation. However, with all these limitations in mind, the project still serves as an excellent base for the future development and implementation of such solutions. The project proves the applicability of the combination of blockchain and machine learning to improve decision-making processes.

1.5 Methodologies of Problem Solving

InfraSync is a proposed solution that uses a methodical and technology-oriented approach to tackle various issues related to the management of road construction projects, including transparency issues, ineffective usage of money, and inaccurate calculation of costs. In implementing this approach, blockchain technology, machine learning, and a website will be integrated.

1. **Blockchain Technology for Transparency and Security**

   The blockchain technology plays an important role in creating a secure and reliable system by providing transparency and security in fund transfer transactions. The records of all such financial transactions will be maintained through the usage of a decentralized ledger which makes it impossible for any kind of fraudulent activity. Through the use of smart contracts that are coded in Solidity language, the process of fund transfer gets automated. There are specific pre-set conditions that are required for the release of funds. Once those conditions are fulfilled, the smart contract releases the fund to the relevant party without any hassle. In this way, blockchain technology provides high reliability and transparency in transactions.

2. **Cost Estimation Using Machine Learning**

   To tackle the challenge of inaccurate cost estimation, the system uses a machine learning algorithm called XGBoost. The model is trained using suitable data and can predict the cost of building structures based on the cost of materials, labor, and the magnitude of the task. The process of training the model requires data preparation, selecting features, and training the model. Once the model is ready, it can be used to estimate the cost of future projects, thus helping officials in their decision-making process. By adopting this data-based method, there will be reduced dependence on traditional approaches to cost estimation, which increases the likelihood of overspending on projects.

3. **Web-based Application for System Interaction**

   A web-based application will be developed, which acts as an interactive interface that will help users like government bodies and contractors. The user interface is designed in such a way that users find it easy to create projects, define milestones, and keep a check on the transaction flow. Business logic, data management, and front-end and back-end interaction will be handled through the back-end process. Interaction with smart contracts using Web3 APIs will help manage interaction with the blockchain network. This will make sure that the users can interact effectively with the system and have access to everything they need in one place.

4. **Workflow of Integrated System**

   The entire methodology works together in a single workflow, which includes: An administrator generates a project and initiates a smart contract. Funding is provided and safely stored in the blockchain. Milestones are set inside the smart contract for the project. The contractor provides status updates of the project through the web portal. Once the milestones are validated, payments are released automatically by the smart contract. Machine learning algorithms help determine costs while planning the project. This workflow of the integrated system allows seamless interaction between the various elements of the system.

   Through the use of blockchain technology, machine learning, and the web-based interface, the above approach manages to counteract all the weaknesses associated with previous approaches. It promotes transparency during financial transactions, enhances the precision of cost estimation, and establishes an organized project management system.

Chapter 2

LITERATURE SURVEY

2.1 Review of Recent Literature

The existing literature is summarized in Table 2.1. A comprehensive review of the literature on the use of machine learning and blockchain technologies for construction management and public finance administration reveals several breakthroughs as well as existing drawbacks. Machine learning is proven to be efficient in predicting construction costs. Machine learning algorithms can handle large amounts of data, detect complex non-linear dependencies between cost-related factors, such as material costs, workforce costs, construction project scale, and time required. Such advanced approaches as XGBoosting and ensemble learning algorithms exhibit more accurate results than conventional statistical approaches. Machine learning algorithms minimize prediction errors, optimize cost estimation, and improve project planning. On the contrary, existing literature mostly focuses on model efficiency and accuracy in terms of prediction. Other aspects of construction project management, such as financial transparency and accountability and monitoring of construction projects' budget allocation, are usually overlooked. As a consequence, machine learning algorithms do not address corruption or fraudulent activities involved in project execution despite improving decision-making at the planning stage. Blockchain technology, on the other hand, is a revolutionary technique aimed at improving transparency and safety during financial transactions and construction. Blockchain allows using distributed ledger systems that record all transactions in a tamper-proof way. Therefore, all financial processes become fully traceable. Moreover, the application of smart contracts contributes to automation of agreements and enforcement of predefined terms without any human interference. Blockchain technology is widely applied in PFM to minimize risks of fraud, data manipulation, and misconduct during tendering, allocation of funds, and supervision of projects. However, blockchain technologies are mostly utilized for transaction management and security. Currently, blockchain tools lack capabilities for predictions and intelligent decision-making. Hence, although blockchain provides users with necessary information about transactions and processes, it does not help to make estimations and improve prediction skills. In recent years, the combination of blockchain with machine learning algorithms attracted much attention. Hybrid systems of such type help integrate predictive power of machine learning algorithms with security provided by blockchain. Machine learning-based blockchain systems allow both forecasting and

Table 2.1: Literature Survey of Existing Approaches

Sr. No.	Title	Methodology	Findings
1	ML-Based Construction Cost Prediction	Applied ML models (XGBoost, Random Forest) on structured datasets	High prediction accuracy but lacks transparency
2	ML Cost Forecasting for Road Infrastructure	ML models used for cost overrun prediction using project parameters	Reduced estimation error but limited to prediction
3	ML with BIM for Construction Management	ML integrated with BIM for planning and delay prediction	Improves planning but lacks financial transparency
4	Blockchain in Public Financial Management	PRISMA-based systematic review of blockchain applications	Ensures transparency, traceability, and fraud prevention
5	Blockchain-Based Government Fund Management	Blockchain used for fund allocation and tracking	Improves accountability but lacks predictive capability
6	Blockchain-Based Construction Tendering	Smart contracts for bid evaluation and contract awarding	Enhances transparency but limited to procurement phase
7	Smart Contracts in Construction Industry	Automation of construction workflows using smart contracts	Reduces disputes but lacks ML integration
8	COPULA: Blockchain Monitoring System	Ethereum-based decentralized project monitoring system	Improves data integrity but lacks prediction
9	Blockchain for E-Governance	Blockchain framework for public fund tracking	Transparent system but lacks domain-specific model
10	Blockchain in Public Finance Systems	Blockchain for tracking government expenditure	Reduces corruption but lacks ML integration
11	ML and Blockchain Integration Review	Review of ML + blockchain across domains	Shows synergy but lacks real implementation
12	AI and Blockchain in Infrastructure	AI + blockchain integration for infrastructure systems	Improves decision-making but lacks system-level design
13	AI & Blockchain in Public Finance	AI for fraud detection and blockchain for tracking	Improves efficiency but not construction-specific
14	ML-Based Road Cost Prediction (India)	ML models applied for highway cost estimation	Accurate prediction but lacks transparency
15	Blockchain Tender and Contract Management	Smart contracts for contract life-cycle management	Improves automation but lacks prediction
16	Integrated ML and Blockchain Systems	Combined ML prediction with blockchain tracking	Addresses integration but limited scope

transaction monitoring. These solutions, however, tend to be quite general in nature and are seldom customized to cater to the needs of the construction sector. They lack important features related to project cycle management, milestone-driven operations, and difficulties encountered during the development of roads and infrastructure facilities. In addition to this, many existing tools fail to offer a comprehensive solution that integrates the components of prediction, funding, and project monitoring.

2.2 Gap Identification/ Common Findings From Literature

From the literature review done above, some of the key observations and research gaps include:

- Machine learning algorithms have proven very efficient in predicting construction costs by analyzing large volumes of data and establishing correlations between factors such as cost of materials, labor, size, and time frame of the project. But most of the research work focuses on making the prediction model more accurate and does not discuss financial transparency and accountability.
- Blockchain technology is commonly applied in providing security and financial transparency in transactions within construction companies and public financial management systems. This increases traceability and automation through smart contracts while reducing fraudulent activities. But these blockchain platforms cannot predict future events.
- Some existing research on how machine learning can be combined with blockchain technology to integrate their advantages proves that this can help achieve both prediction and financial transparency. But these kinds of applications tend to be general and not specific to construction or road infrastructure.
- Many of the existing applications tend to concentrate on one area, whether cost prediction, fund tracking, contract management, among others.
- There is an absence of comprehensive systems for managing projects throughout their entire lifecycle, including their execution based on milestones, budgeting, monitoring, and evaluation through one integrated system.
- Existing methods are unable to cope with practical issues arising in road construction, including dynamic project conditions, broad project implementation, and system integration.

Chapter 3

SOFTWARE REQUIREMENTS SPECIFICATIONS

3.1 Functional Requirements

Functional requirements state the essential actions and functionalities of the InfraSync system that will make it possible for it to serve the users' needs and reach its goals. In other words, functional requirements will determine how the InfraSync system will function depending on certain criteria. It should be noted that the InfraSync system is designed in such a way that it will be possible to achieve the desired results through the use of blockchain technology, machine learning, and web interface.

3.1.1 System Feature 1: Blockchain-based Fund Allocation

A feature offered by the system is that of allocating the funds using blockchain technology. This helps improve accountability among all the stakeholders in the process. Government bodies are able to allocate funds to specific road constructions through smart contracts running on the blockchain platform.

Functional Requirements:

- The system should provide a way to deploy and develop smart contracts to govern fund allocations.
- The system should keep track of all financial transactions.
- The system should provide immutability, meaning that there will be no modification to the transaction records once they have been recorded.
- The system should provide transparency by enabling the stakeholders to view transactions such as the amounts, transaction time, and hashes.
- The system should provide a secure means to handle all the transactions using the capabilities of the blockchain platform.

3.1.2 System Feature 2: Milestone-based Payment System

The milestone-based payment system ensures that money can only be withdrawn after the completion of certain milestones within the project. This guarantees transparency and

effective usage of the funds. The project is broken down into various milestones, and payments are tied to the successful completion and validation of these milestones.

Functional Requirements:

- The system should enable milestone definition when setting up a project.
- The system should tie a certain amount of money to each milestone.
- The system should validate and release money only after successful validation of the milestone.
- The system should block any unapproved withdrawals of the money.
- The system should log all milestone payments to the blockchain.
- The system should inform users about milestone payments.

3.1.3 System Feature 3: Predicting Construction Cost using Machine Learning

This system will integrate a machine learning component that predicts the construction cost of projects depending on the input parameters. This will help facilitate effective planning and budgeting of construction projects. The machine learning algorithm used is the XGBoost to predict construction costs.

Functional Requirements:

- The system should receive inputs of project parameters such as cost of materials, amount of labor, and scale of project among others.
- The system should pre-process the inputs for more accurate prediction of construction costs.
- The system should apply the XGBoost algorithm to predict the cost of construction.
- The system should provide output estimations of the costs of construction.
- The system should help planning and budgeting through the provided estimates.

3.1.4 System Feature 4: User Management System

The system incorporates a user management system that takes care of issues like user authentication, authorization, and access control. Various user roles are created to cater to the requirements of responsibility management.

Functional Requirements:

- The system shall enable users to create accounts and sign in.

- The system shall use user authentication based on credentials.
- The system shall include user roles like Admin (government authority) and contractor.
- The system shall employ user access control based on user roles.
- The system shall create logs of user activities.
- The system shall enable users to edit their profile details.

3.1.5 System Feature 5: Project Management System

It ensures that users have full functionalities for handling road construction projects from conception till completion. It is designed to help in tracking and monitoring project activities. This is one of the most critical features which serves as the backbone of the system.

Functional Requirements:

- It should give administrators the ability to add new road construction projects into the system.
- It should store information relating to project like project names, budget, timelines, and milestones.
- It should give contractors the chance to update progress in the projects.
- It should give authorities the chance to check the progress of projects using a dashboard interface.
- It should update the progress of the projects in real time.

3.2 External Interface Requirements

The external interface requirements indicate the interactions that need to take place between the InfraSync system and users, external hardware, software, and communication protocols. These will ensure that the system integrates well with external entities and interacts effectively.

3.2.1 User Interfaces

InfraSync user interface comes with a web-based application that is intended to be user-friendly and intuitive. The user interface helps the different types of users including

government authority (Admin) and contractors to interact with the platform efficiently. Modern frontend technologies have been used in designing the interface for InfraSync with attention to creating a user-friendly navigable design and accessible UI.

Key features of InfraSync user interface include:

Dashboard for Government Authorities A view of all projects and their fund allocation status. Admin can easily create projects and track the progress of the projects.
Contractor Interface Helps the contractor see their assigned projects and report on project progress including milestone completion.
Project Visualization Project details including the budget, timeline, milestones, and project status can be visualized from this section.
View Transaction History A view of transaction history which contains information about the transaction hash, transaction amount, and timestamp.
Simple design User-friendly and straightforward designs make interacting with the interface effortless.

3.2.2 Hardware Interfaces

No special hardware needs to be installed in the InfraSync architecture.

3.2.3 Software Interfaces

The system integrates with a variety of software elements in order to operate optimally. The following interfaces help the system in communication. Software interfaces are as follows: Blockchain Network Platform (Ethereum or EVM-based): Utilized to deploy smart contracts for managing funds allocation and transactions.

- **MetaMask Wallet:** Serves as an interface for the user to communicate with the blockchain network. Authentication and secure blockchain transaction signing are done using MetaMask.
- **Backend Server (Node.js or Express.js):** Involved in implementing business logic, processing API requests and communication between frontend and database.
- **Frontend Framework (Next.js):** Used to create the graphical user interface of the application.

All these software interfaces integrate seamlessly to provide functionality of all system modules.

3.2.4 Communication Interfaces

The interface communication defines the rules for the exchange of information between different components of the system. The communication techniques used for the system include:

- **HTTP/HTTPS Protocol:** For secure communication between the client (front end) and the server (back end).
- **REST APIs:** Allows interaction between front and back end through requests for creating a project, retrieving data, and updating data.
- **Web3 APIs:** Provides communication between the application and the blockchain network to perform smart contract actions and retrieve transactions.

Communication interfaces guarantee efficient data exchange and responsiveness of the entire system.

3.3 Non-Functional Requirements

Non-functional requirements represent the quality characteristics and performance criteria of the InfraSync system. They guarantee the smooth operation of the system in various scenarios. They emphasize the manner of performing tasks rather than the actual task performed by the system itself.

3.3.1 Performance Requirements

Performance requirements determine the effectiveness of the system from the perspective of its speed, responsiveness, and scalability. It is expected that the system will allow many users to work with it at once with no noticeable decrease in performance. It will allow both administrators and contractors to use it at once. Transactions within the blockchain should be performed quickly and efficiently without delays in their processing and confirmation. Though blockchain might cause some delay due to its nature, this problem should be addressed within the system's functionality. It is required that all interactions between the system and user (login to the system, creating a project, requesting information, and other operations) will be performed quickly to improve usability. The system should guarantee efficient interaction between its frontend and backend part, thus allowing fast data transmission from one to another. The system should be scalable, meaning that it will be able to cope with the growing number of users and projects in the future. The system should ensure consistency and reliability of its operation, meaning no delays or disruptions while providing a service.

3.3.2 Safety and Security Requirements

Safety and security requirements make sure that the system ensures safety for the stored information, preventing any unauthorized access, and maintaining the system's integrity. All operations related to money must be encrypted using blockchain cryptography algorithms to ensure data integrity and prevent any alterations. User login into the system should be done via reliable mechanisms, making sure that the system is used by authorized personnel only. Data integrity of the stored data should be maintained in order to provide consistency in the system. Access to the system features should be provided according to the roles assigned to users, such as Admin or Contractor, preventing unauthorized access to the system. Sensitive data should be stored in a secure manner, applying encryption or hash functions where possible. The system should guarantee secure data transfer between its components through the use of protocols like HTTPS.

3.4 System Requirements

3.4.1 Database Requirements

User information, project information, and feedback (if any) shall be stored by the system. MongoDB may be used for the database. Transaction information shall be stored by Blockchain.

3.4.2 Software Requirements (Platform Choice)

- Operating System: Windows/Linux
- Frontend: Next.js
- Backend: Node.js / Express.js
- Blockchain: Solidity (Ethereum/EVM)
- Tools: MetaMask, Hardhat

3.4.3 Hardware Requirements

- Minimum 8 GB RAM
- Processor: Intel i5 or equivalent
- Internet connectivity

3.5 SDLC Model to Adopt

The software project will be developed following the Agile Software Development Life Cycle (SDLC) model. The development process will be done iteratively. Modules like integration of blockchain technology, backend, and frontend will be built in a step-by-step manner. Agile approach is flexible.

Chapter 4

SYSTEM DESIGN

4.1 System Architecture

InfraSync system architecture is proposed as a multi-level and modular design combining the use of blockchain technology, machine learning, and web application (Fig 4.1). The architecture seeks to enhance transparency in fund distribution, manage projects efficiently, and estimate costs accurately while maintaining scalability, security, and usability. Each layer of the architecture has its function, and communication takes place between all the layers for the functionality of the system.

1. **User layer:** This layer contains users who are the final users of the system and access the system via web interfaces. There are two major types of users supported by the system:

    - **Government Authorities (Admin):** They have the responsibility of creating projects and funding them, determining the milestones for each project, and monitoring the progress of each project.
    - **Contractors:** They are tasked with updating the progress of the projects and achieving the milestones set for each project. It is an entry portal that allows users of different types to use the system.

2. **Frontend Layer** It uses Next.js and provides an interactive interface for the users. It is responsible for visual presentation of the data and interaction with the system by the users. Functionalities of the frontend layer include:

    - Providing dashboard for project summary and transaction records
    - Form creation for project and milestones creation
    - Ability to view blockchain transaction history
    - Updating project status by the contractors

    It interacts with the backend using REST APIs.

3. **Backend Layer** Backend is developed using Node.js and Express.js frameworks and forms the core processing layer of the system. This layer deals with various tasks related to business logic and data processing. Key functionalities of the backend layer include:

- User request processing received from the frontend layer
- Data management related to projects and users
- Database interaction for data storing/retrieval operations
- Interaction with the blockchain layer using Web3 APIs
- Integration with the machine learning module for cost prediction

Thus, this layer connects frontend, blockchain and machine learning modules in one place.

4. **Blockchain layer:** Blockchain is an important part of the system that makes sure that there is no fraud regarding any financial transaction made through this platform. Smart contracts have been developed using Solidity and deployed in the Ethereum/EVM based environment. Blockchain layer is used to perform some important tasks like:

   - Fund management for various projects
   - Milestone-based payments
   - Storing all transactions in an immutable ledger form
   - Data security and integrity management

5. Machine Learning Module Machine Learning module plays an important role in determining the cost of road construction projects. The module employs the use of XGBoost algorithm for analyzing input parameters such as material prices, labor needs, and project size. Some of the key functions of the module include:

   - Analyzing input data to estimate project cost
   - Providing predictions regarding the project cost
   - Helping authority to plan and budget

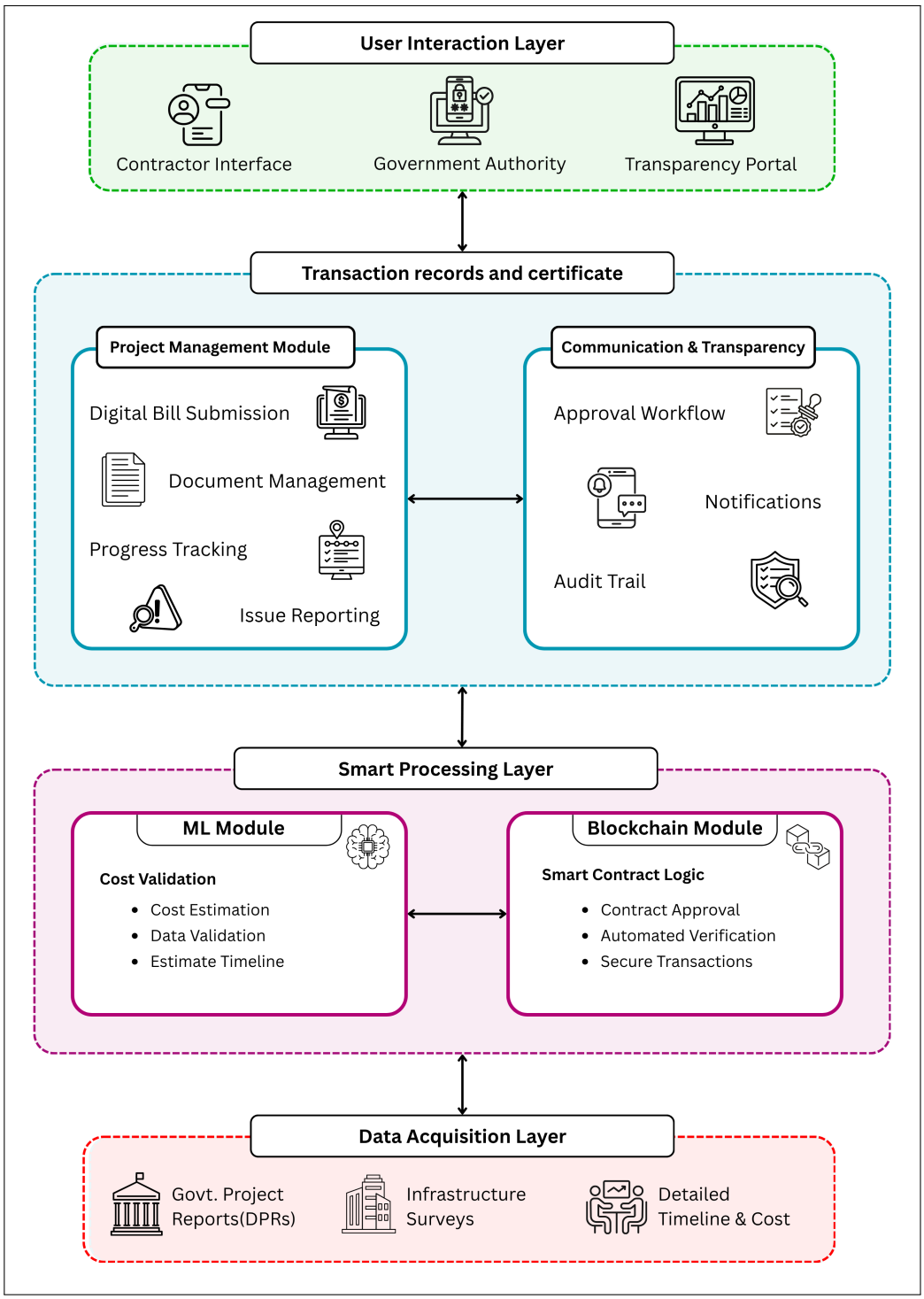


Figure 4.1: System Architecture of the Proposed Model

System working flow:

The general functioning of the whole system may be explained as a set of operations done at different levels:

1. Admin signs up for the system and adds new projects to the database.
2. Smart contract is created for the project on the blockchain.

3. Money is raised for the project and put into the smart contract.
4. Project milestones are specified upon initialization.
5. Contractor reports progress on the project via the web interface.
6. On completion of the milestones, the smart contract pays out funds
7. All transaction details are recorded on the blockchain and made visible to all participants
8. Machine learning module predicts the cost of the project.

The architectural model proposed will make sure that there is a clear distinction of roles between various modules, thus allowing seamless operations and scalability of the system. Blockchain technology makes sure that there is clarity and security in the process while machine learning helps in making decisions. As a result, there will be an effective architecture for management of road construction projects.

4.2 Data Set / Database Structure

The InfraSync system adopts a combined data storage strategy whereby a conventional database handles application-related data, whereas blockchain technology handles financial transaction data. This allows for efficient and transparent management of data. MongoDB, which is a NoSQL database, is employed to handle the database structure. This database offers flexibility and scalability and enables effective management of structured and semi-structured data. Collections in MongoDB are used to categorize data according to the requirements of the system.

4.2.1 Database Design (MongoDB)

Following collections are utilized in the system:

1. Users Collection This collection contains data for all registered users in the system. Fields:
    - User ID: User's unique identification number
    - Name: Name of the user
    - Role: Determines the role of user (Admin / Contractor)
    - Email: Identification for logging in and contacting the user

- Password: Encrypted password for user's security

Function:

- Used for managing the login credentials of the users
- Provides authorization and authentication for the system

2. Projects Collection Collection containing data related to road construction projects. Fields:

- Project ID: Unique number to identify the project
- Project Name: Project's name
- Description: Description of project
- Budget: Fund allotted for project
- Time-line: Length of time period needed for completion
- Status: Current status of the project (ongoing, completed)

Function:

- Maintains data of all the projects
- Provides the facility for monitoring and tracking the progress of projects

4.2.2 Blockchain Storage

The blockchain layer provides data security by storing the crucial financial and contract-related data. Data stored on blockchain include

- Fund Allocation Records: Records detailing the allocation of funds to each project
- Milestone Payments: Payment records after milestone completion
- Smart Contract Information: Contract address and other details regarding execution conditions

Purpose:

- Provides transparency
- Prevents alteration of records
- Provides an unchangeable record of transactions

4.2.3 Dataset for ML model:

The creation of dataset involved inclusion of parameters that would affect road construction costs, based on their real-life occurrence and estimation of infrastructure projects. Parameters included in the dataset include:

- Length of road construction (in km)
- Cost of materials
- Labor cost
- Total number of laborers
- Construction period (in days)
- Earthwork volume (in m3)
- Height of embankments (in m)
- Total number of bridges
- Total number of culverts
- Drainage work length (in m)
- Average salary/day

The above parameters form part of each row in the dataset, along with the output value representing cost of construction of the particular infrastructure.

4.3 Mathematical Model

The InfraSync system's mathematical model employs a supervised learning algorithm for predicting the cost of constructing road networks. The XGBoost (Extreme Gradient Boosting) learning algorithm is employed, which is an ensemble method that makes use of several decision trees for making predictions.

4.3.1 Input and Output Representation

Let the input features be represented as:

$$X = \{x_1, x_2, x_3, \ldots, x_n\} \tag{4.1}$$

where each feature x_i corresponds to relevant parameters affecting construction cost, such as material cost, labor cost, project size, project duration, and other influencing factors such as:

- Material cost
- Labor cost
- Project size
- Duration of the project

The output variable is defined as:

$$Y = \text{Predicted Construction Cost} \tag{4.2}$$

4.3.2 Prediction Function:

The XGBoost model predicts the output by combining the results of multiple decision trees. The prediction function is expressed as:

$$Y = \sum_{k=1}^{K} f_k(X) \tag{4.3}$$

where:

- $f_k(X)$ represents the prediction made by the kth decision tree,
- K is the total number of trees in the model.
- Each tree contributes to improving the overall prediction

The final predicted cost is obtained by summing the outputs of all individual trees.

4.3.3 Objective Function

The objective of the model is to minimize the prediction error between the actual cost and the predicted cost. This is achieved using a loss function, typically defined as:

$$\text{Loss} = \sum (Y_{\text{actual}} - Y_{\text{predicted}})^2 \tag{4.4}$$

where:

- Y_{actual} is the actual construction cost,
- $Y_{\text{predicted}}$ is the cost predicted by the model.

This loss function ensures that the model learns to reduce the difference between actual and predicted values during training.

4.3.4 Model Optimization

Model optimization through XGBoost is achieved through an iteration technique where each subsequent tree is optimized in such a way that it corrects the errors made by the preceding trees. This method is called gradient boosting and helps in:

- Reducing errors and variance
- Improving accuracy of predictions
- Managing the complicated relationships between variables

4.3.5 Importance of the Model

In the InfraSync system, the mathematical model helps in:

- Accurate estimation of costs incurred in construction processes
- Making decisions based on data analytics
- Decreasing dependence on manually-based cost estimation techniques

By employing the mathematical model based on XGBoost, the system can produce cost predictions with high levels of accuracy and reliability.

4.4 UML Diagrams

4.4.1 Use case diagram

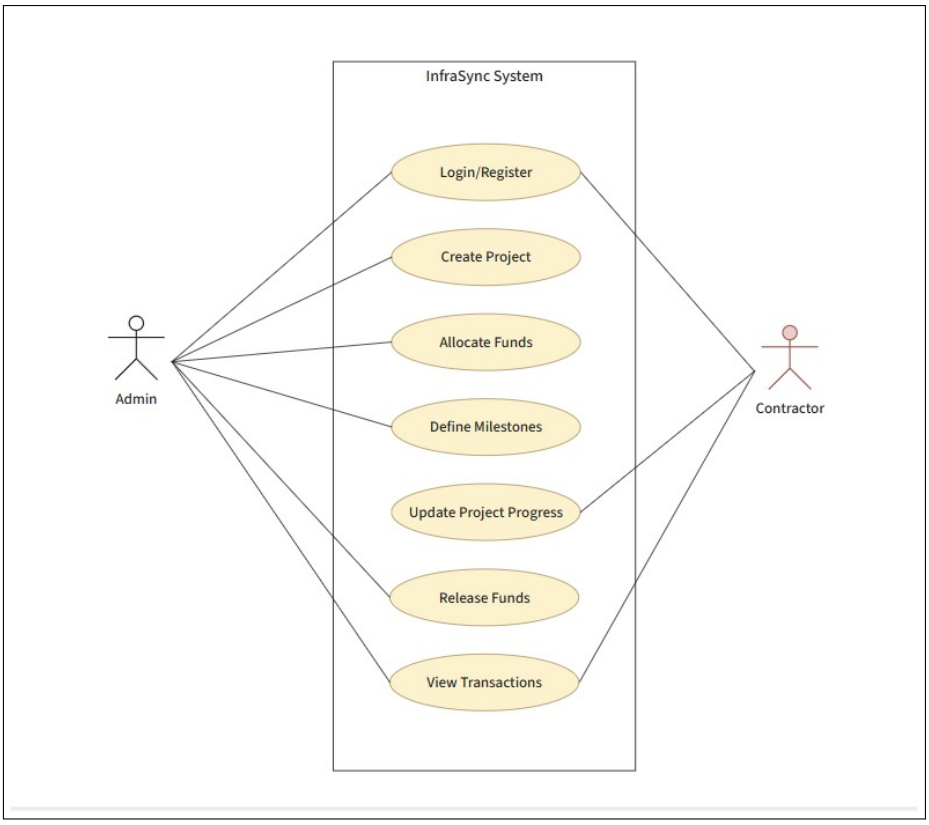


Figure 4.2: Use Case Diagram

The use case diagram (Fig. 4.2) represents interactions between users and the system.

Actors:

- Admin (Government Authority)
- Contractor

Use cases:

- Login/Register
- Create Project
- Allocate Funds
- Define Milestones
- Update Project Progress

- Release Funds
- View Transactions

4.4.2 Class Diagram

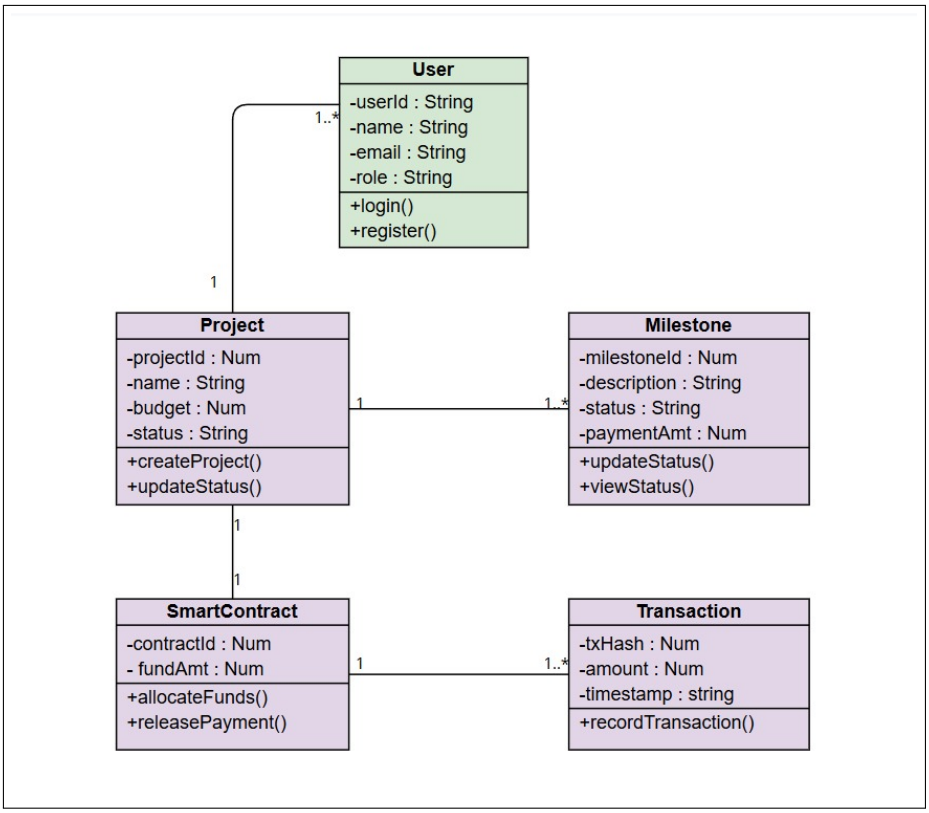


Figure 4.3: Class Diagram

The class diagram (Fig. 4.3) defines the structure of the system.

Main classes:

- User
- Project
- Milestone
- Smart Contract
- Transaction

Attributes:

- IDs, names, roles, budget, status, etc.

Methods include:

- createProject()
- allocateFunds()
- releasePayment()
- updateStatus()

4.4.3 Sequence Diagram

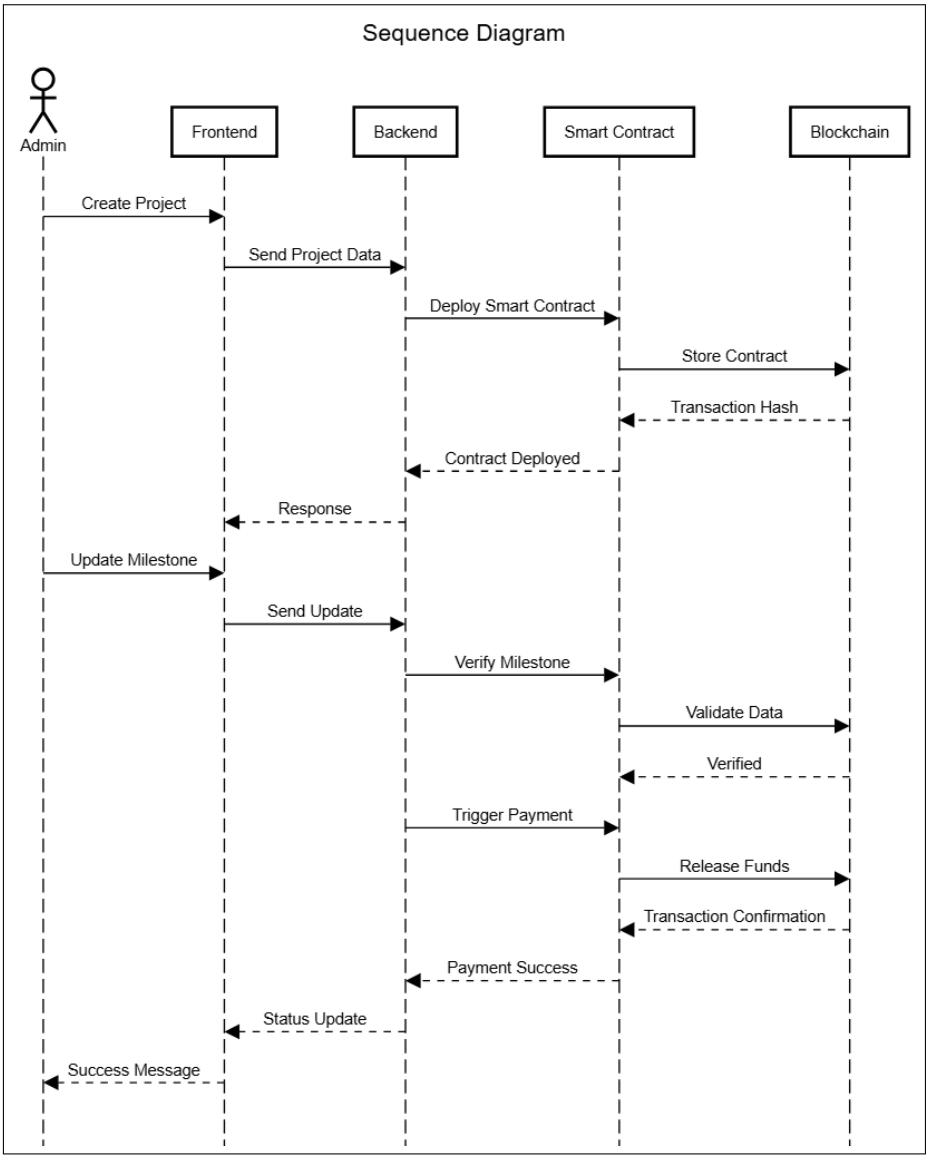


Figure 4.4: Sequence Diagram

The sequence diagram (Fig. 4.4) shows the flow of operations in the system.

4.4.4 Activity Diagram

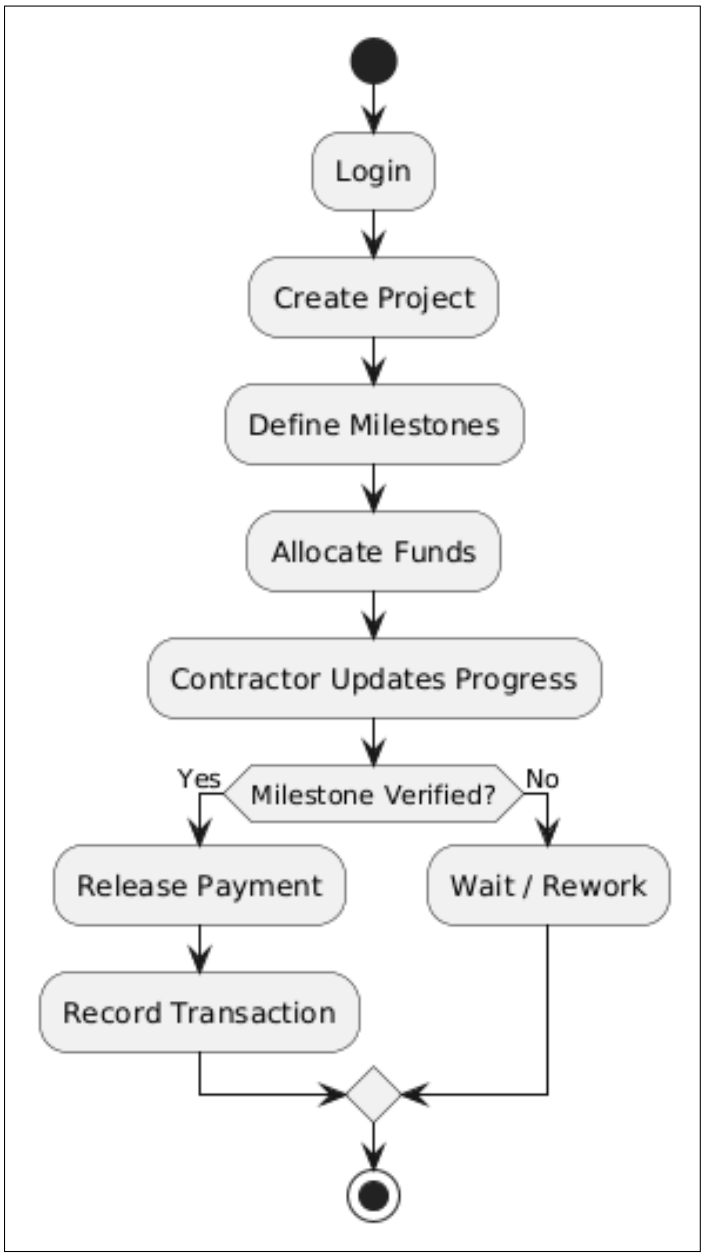


Figure 4.5: Activity Diagram

The activity diagram (Fig. 4.5) represents workflow of the system.

Flow includes:

- Project creation
- Fund allocation

- Milestone tracking
- Verification
- Payment release

Chapter 5

PROJECT PLAN

5.1 Risk Management

Risk management includes identification, analysis of potential threats, and planning actions for minimizing them.

5.1.1 Risks Identification

The following risks have been identified during the project development:

- Difficulty of Blockchain Integration: Problems of deployment and interaction with smart contract.
- Transaction Delay: Time consumption of transactions' confirmation on blockchain.
- Incorrect Input Data: Mistakes in providing data for predicting costs.
- System Integration Risks: Issues in integration between frontend, backend, and blockchain.
- Errors of Users: Wrong usage of the system by users.

5.1.2 Risks Analysis

Risks related to blockchain technologies may affect system reliability and performance.

- Transaction delays might decrease user experience.
- Incorrect data may cause mistakes in cost predictions.
- Risks related to integration process may delay project development.

Every potential risk has been assessed for probability and possible consequences to the system.

5.1.3 Risks Mitigation, Monitoring, and Management

- Using Tested Frameworks: Special frameworks (e.g. Hardhat) have been used for simplification.

- Testing Environment: Smart contracts were tested on test/local networks before deployment.
- Validation of Data Input: Proper validation of input data was implemented.
- Modular Approach: Components of the system were developed independently.
- Monitoring and Continuous Testing: Testing and debugging were performed regularly.

5.2 Project Schedule

5.2.1 Set of Project Tasks

The project involved breaking down the tasks into phases for systematic execution:

- Identification of Problem and Requirement Analysis
- Survey of Literature
- System Design
- Development of Blockchain Module
- Development of Machine Learning Module
- Development of Web Application Interface (Frontend & Backend)
- Integration of System Elements
- Testing and Debugging
- Documentation

5.2.2 Timeline Graph

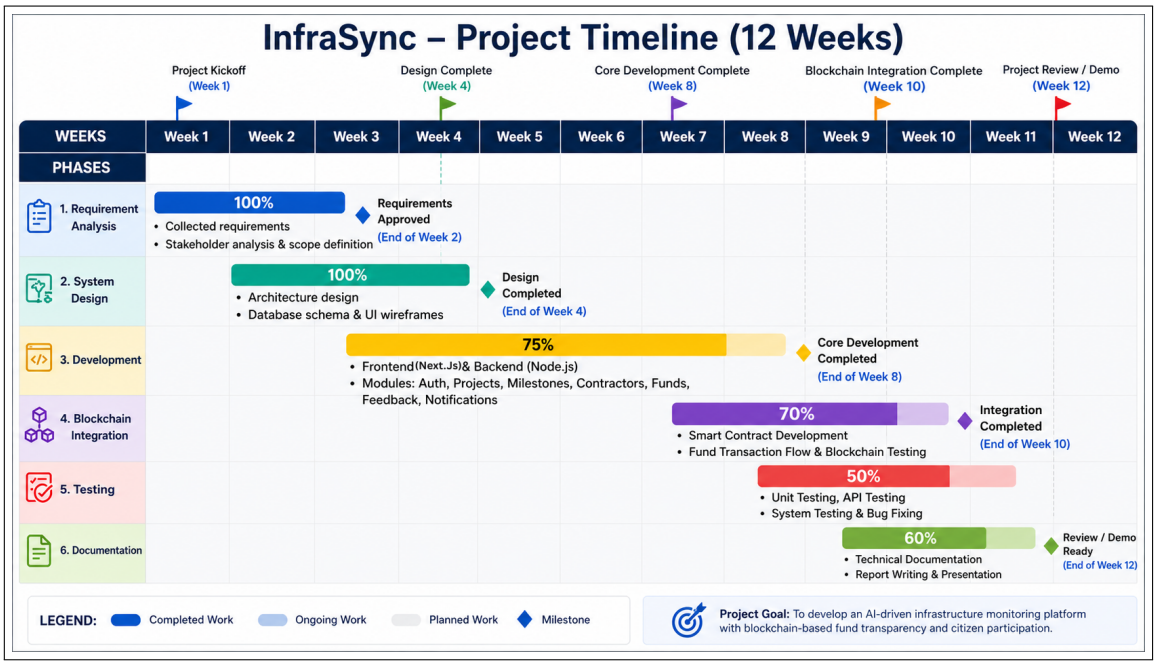


Figure 5.1: Timeline Graph

The project was executed within a pre-defined timeline with each phase assigned a particular amount of time. Fig. 5.1 shows the timeline graph of the development process.

- Early phases including analysis and design were executed first.
- Later phases were developed in parallel wherever possible.
- Phases related to integration, testing, and documentation followed.

5.3 Team Structure

The project was created using a well-defined team with specific tasks and roles.

Roles within the Team:

- **Team Leader: Om Sagar**
  Tasked with coordinating the entire project, assigning tasks, and monitoring the project's progress.
- **Blockchain Developer: Vinayak Suryawanshi**
  Tasked with coding the smart contract, deploying it, and testing it.

- **Backend Developer: Om Sagar**
  Tasked with coding the backend logic of the application.
- **Frontend Developer: Parag Pawar**
  Tasked with coding the frontend design of the application.
- **Machine Learning Developer: Harshal Suroshe and Vinayak Suryawanshi**
  Tasked with creating the cost prediction algorithm.

Team Collaboration:

- Team meetings were held regularly to monitor the project's progress.
- Tasks were allocated based on the skills of the members.
- Collaboration resulted in the successful completion of all modules.

Chapter 6

PROJECT IMPLEMENTATION

6.1 Overview of Project Modules

InfraSync system has been built using a modular architecture design wherein the entire system has been divided into a number of modules. Every module will take care of certain functions, thereby making the system more organized and scalable.

These are some of the important modules that make up our system:

1. **User Management Module** The User Management Module deals with all those processes related to the user management. The module takes care of registering the users and also manages their actions.
   Functions associated with the module include:

   - Registration of users using their credentials
   - Authentication of the user from the database
   - Role-based access rights for various types of users
   - User profile management with user details
   - Activity logs of the user

   Our system supports various user types with these primary ones:

   - Admin (Government Authority): Manage the projects and funds
   - Contractor: Updates project status

   This module acts as a security component of our system and restricts access to functionalities by various users.

2. **Project Management Module** The Project Management Module is designed to conduct all operations associated with road construction projects. This module serves as the main component where project data is created and stored.
   Main features include:

   - Generation of road construction projects by the admin

- Data storage of the project, including its name, budget, deadline, and current state
- Milestones definition and management
- Project update by contractors
- Project tracking by the authorities

All project-related data is collected and organized into one place via this module to help users manage their projects efficiently.

3. **Blockchain Module** The Blockchain Module is an essential part of our system and performs functions associated with the secure and immutable financial transaction management process. The module uses the technology of smart contracts written in Solidity and deployed on an Ethereum/EVM-like platform.
   Main features include:

   - Smart contract deployment for each project
   - Project funding through allocating necessary amounts of money to it
   - Milestone-based payments processing
   - Transaction recording onto the blockchain ledger
   - Prevention of any changes to transaction data
   - This feature will eliminate any centralized control over the financial transaction process.

4. **Machine Learning Module** Machine learning module helps in estimating cost for road construction project predictions. This tool can assist in making accurate cost estimates for proper planning and decision making.
   Key features of ML module include:

   - Receiving input parameters such as cost of materials, labor cost, and project cost
   - Data processing for the purpose of cost prediction
   - Using XGBoost algorithm for generating cost predictions
   - Sending output predictions to users
   - This feature reduces dependence on estimation based on past experiences.

6.2 Tools and Technologies Used

The system is developed using modern technologies for efficient implementation.
Frontend Technologies:

- Next.js – for building user interface
- HTML, CSS – for structure and styling

Backend Technologies:

- Node.js – for server-side logic
- Express.js – for API development

Blockchain Technologies:

- Solidity – for writing smart contracts
- MetaMask – for transaction handling
- Hardhat – for development and testing

Machine Learning:

- Python – for model development
- XGBoost – for cost prediction

Database:

- MongoDB – for storing application data

6.3 Detailed Description of Algorithms

InfraSync uses several algorithms which describe the sequence of actions for executing critical functionalities such as fund allocation, cost estimation, user authentication, and management of projects. The use of algorithms guarantees the operation of the software in an organized, secure, and efficient way.

6.3.1 Algorithm 1: Blockchain-based Fund Allocation and Payment

Description:
This algorithm describes the procedure of distributing funds and making payments by utilizing blockchain technology and smart contracts. Payments can only be done following the successful completion of certain predefined milestones.

Steps:

1. Administrator logs into the application.
2. Administrator creates a project in the system by specifying the necessary details about the project.
3. A smart contract is deployed on the blockchain network.
4. Funds are allocated to the smart contract for the project.
5. Milestones for the project are established and added to the smart contract.
6. Contractor specifies the completion of a milestone.
7. System checks the completion of the milestone.
8. If milestone verification was successful: Smart contract releases the payment.
9. Updated transaction information is shown to the users.

Result:
Automated and secure distribution of funds.

6.3.2 Algorithm 2: Predict Estimated Cost using Machine Learning

Description:
This algorithm uses an existing XGBoost machine learning model to estimate the cost of construction.
Algorithm Steps:

1. Collect the input values (e.g., material cost, labor cost, project size).
2. Preprocess data (i.e., cleaning, normalizing, and formatting).
3. Load the pre-trained XGBoost model.
4. Provide input data to the model.

5. The model will process the input data using multiple decision trees.
6. Each tree will make partial predictions.
7. Combine the output of each tree to give the final output.
8. Show the construction cost prediction result to the user.

Outcome:
Accurate data-driven cost prediction.

6.3.3 Algorithm 3: Authenticate User and Manage Roles

Description:
This algorithm controls the access of the users and implements role-based permission control inside the application.
Algorithm Steps:

1. The user logs in using their email address and password.
2. The system collects the information related to the user.
3. The system authenticates the credentials provided by the user.
4. If the credentials are correct: The system successfully authenticates the user.
5. The system determines the role of the user (Admin/Contractor).
6. Assign the access rights according to the role.
7. Redirect the user to the respective dashboard.

Outcome:
Secure authentication and role-based access control.

6.3.4 Algorithm 4: Project Creation and Monitoring

Description:
The algorithm handles the creation, modification, and monitoring of construction projects.
Algorithm Steps:

1. Admin signs into the system.
2. Admin inputs information on the construction project such as its name, budget, and schedule.

3. Project information is stored in the database by the system.
4. Admin specifies the milestones for the project.
5. Contractor gains access to the assigned project.
6. Contractor reports the progress made on the project and milestone status.
7. Status of the project is updated immediately in real-time.
8. Admin monitors the project through the dashboard.

Output:
Facilitates effective tracking, managing, and monitoring of construction projects.

In summary, these algorithms provide the basic framework of the InfraSync system. They guarantee that tasks like fund distribution, cost estimation, verification, and project management are carried out in an orderly fashion. The utilization of blockchain technology and machine learning boosts the performance of the InfraSync system.

Chapter 7

SOFTWARE TESTING

7.1 Types of Testing

Testing of software applications is a significant step in software development that guarantees that the application performs well, works according to specifications, and is without errors.
In order to make sure the InfraSync system functions perfectly, various types of testing have been performed. Each test concentrates on a unique feature of the application, guaranteeing its correctness and reliability.

1. Unit Testing
   Unit testing refers to the testing process where every module of the system is tested individually for accuracy.

   - Modules like user authentication, creation of projects, and the functioning of smart contracts were tested individually.
   - Valid and invalid inputs were used to test the functions.
   - Any problems found were solved during the early stages of development.

   This test makes sure that each function is accurate in itself.

2. Integration Testing
   Integration testing concentrates on validating the interactivity of the various parts of the system.

   - Tested communication among the frontend (Next), backend (Node.js), and blockchain levels.
   - Validated the proper flow of information across REST and Web3 APIs.
   - Checked whether information from the frontend gets properly processed by the backend and recorded in the form of blockchain transactions.

   These tests make sure that all parts interact smoothly without any issues.

3. System Testing
   System testing means checking the complete system after integrating all parts.

- Entirely tested the InfraSync application in the context of practical usage.
- Checked features like creating projects, allocating funds, and managing milestones.

This test makes sure that all parts function together as intended.

4. Functional Testing
   Functional testing validates whether all the functionalities of the system are performed according to their specifications outlined in the Software Requirements Specification document. Some of the functionalities that were tested include:

   - Fund allocation using blockchain technology
   - Payment disbursement based on milestones
   - Prediction of costs through machine learning techniques
   - Project creation and management
   - Validation of expected outcomes against actual system outcomes.

5. User Acceptance Testing (UAT)
   User Acceptance Testing aims to check the system from the perspective of the users themselves.

   - Usability and ease of use of the system have been checked.
   - It was ensured that both Admin and Contractor panels were intuitive to use.
   - It was verified that the system would meet the needs of users and work properly.

   This testing helps make sure that the system will be used in practice.

6. Security Testing
   Security testing is required to ensure that the system is not vulnerable to unauthorized access or manipulation of data.

   - It was ensured that there was secure authentication in the system.
   - It was also checked that there was no risk to handling of sensitive information like logins.
   - Blockchain transactions immutability was verified.
   - Unauthorized access behavior in the system was tested.

It was thus ensured that the system will protect data integrity.

Use of different kinds of tests guarantees the reliability and safety of InfraSync. Through comprehensive testing of separate modules, as well as testing the system as a whole, errors were detected and solved; this led to stable and functional software.

7.2 Test Cases and Test Results

In order to verify the proper functioning of the InfraSync application and its adherence to all specifications, a series of tests has been carried out.It is in table 7.1. Each test case assesses the performance of a certain functionality in the software, which is done through comparison between the predicted outcome and the result generated during the testing. The test cases are important in determining system efficiency and reliability.

Table 7.1: Test Cases for System Validation

Test Case ID	Test Description	Input	Expected Output	Actual Output	Status
TC01	User Login	Valid credentials	User is logged in successfully	Successful login	Pass
TC02	User Login	Invalid credentials	Error message displayed	Error message displayed	Pass
TC03	Create Project	Valid project details	Project is created successfully	Project created successfully	Pass
TC04	Allocate Funds	Valid fund amount	Funds allocated to smart contract	Funds allocated successfully	Pass
TC05	Deploy Smart Contract	Valid contract inputs	Smart contract deployed successfully	Contract deployed successfully	Pass
TC06	Milestone Completion	Valid milestone update	Milestone status updated	Status updated successfully	Pass
TC07	Fund Release	Completed milestone	Payment released	Payment released successfully	Pass
TC08	Cost Prediction	Valid input parameters	Estimated cost generated	Correct prediction displayed	Pass
TC09	Unauthorized Access	Invalid role access	Access denied	Access denied	Pass
TC10	View Transactions	Valid request	Transaction history displayed	Transaction history displayed	Pass

Result Analysis:

- All identified test cases have been executed successfully and the results obtained matched the anticipated outcomes in all situations.
- Functions such as user identification, management of the projects, fund distribution, implementation of smart contracts, and the estimation of costs were confirmed and validated.
- Operations related to blockchain technology like transaction tracking and the release of funds worked perfectly fine with no anomalies.
- The system exhibited stability and efficient handling of operations throughout the test.

Result Overview:

- The InfraSync prototype was successfully tested and validated against all defined test cases.
- No significant problems were detected in the operation of the system during testing.
- Any minor problems identified during the development process were addressed during debugging and testing.
- The system was operational within the anticipated scope of use, delivering accurate outputs for all test cases.
- The successful execution of all test cases demonstrates that the InfraSync prototype is efficient, reliable, and ready for deployment.

Chapter 8

RESULTS AND DISCUSSIONS

8.1 Findings

InfraSync has been designed, built, and developed to solve the major problems associated with the construction management of roads such as lack of transparency, ineffective allocation of funds, and inaccurate cost estimation. With its integration of blockchain technology and machine learning, InfraSync becomes an advanced and effective system that can manage infrastructure projects.
The key findings of the research are listed below:

1. **Development of Blockchain-based Fund Management System**
   Blockchain-based fund management system was created, which allows carrying out financial operations with ease and security. Any transaction conducted in regards to fund allocation and payments is added to the decentralized ledger, making any change impossible.
   Thus, the system contributes greatly to increasing transparency and trust among parties involved.

2. **Automated Payment System Using Smart Contracts** Smart contracts were developed and implemented on an Ethereum/EVM-based network for automating fund disbursements.

   - The payment system is connected with the milestones set in the initial contract.
   - Payment transactions will only take place upon successfully verifying the accomplishment of the milestone.
   - Corruption and human error in payment transactions are reduced.

   This results in payment transactions being executed efficiently and in accordance with defined rules.

3. **Construction Cost Prediction Model Based on Machine Learning Techniques**
   An XGBoost-based machine learning prediction model was created to forecast construction costs effectively.

- Parameters include the cost of materials, manpower, and project size among others.
- Predictions made by the machine learning model are highly accurate and reliable.
- They help in proper planning and budgeting of construction projects.

The outcome leads to better resource planning and decision-making using data.

4. **Development of Web-based System for User Interface**

   A fully functional web-based application was created to facilitate interaction between users and the system.

   - System admin users can create, edit, distribute funds, and monitor project progress.
   - Contractors can update project statuses and achievements.
   - Both the admin and contractors can track transaction history and project progress.

   It results in easy interaction between the users and the system.

5. **Implementation of Role-based Access Control**
   Access to functionalities is regulated using the role-based access control method, ensuring that access is secure and controlled.There are different types of users, each with unique responsibilities and permissions.

   - For instance, admins are responsible for managing projects and funding.
   - On the other hand, contractors only have the right to update project progress.

   This method limits what unauthorized users can do within the system.

6. **Successful Integration of System Components**
   The components have been successfully integrated into the system.
   These include:

   - Frontend (Next.js)
   - Backend (Node.js/Express.js)
   - Blockchain (Smart Contracts)
   - Machine Learning Model

7. **Demonstration of Feasibility and Practical Implementation**
   The application has demonstrated the possibility of utilizing new technology, such as blockchain and machine learning models, in infrastructure management.

   - The system works as a prototype.
   - This proves that the concept is practical and can be applied in real life.

   Overall, the InfraSync system has successfully met its intended objectives, which are to provide an automated and efficient means of managing road construction projects.

8.2 Results Analysis and Validations

The evaluation of the performance and efficacy of the InfraSync framework considered several factors, such as functionality, precision, reliability, and system behavior. The assessment process involved the analysis of individual components and the whole integrated system. It is evident from the results obtained that the system operates effectively and attains the desired outcomes.

1. **Performance Evaluation of ML model**
   The algorithm used in the system to predict the cost of road construction includes XGBoost (Extreme Gradient Boosting). The training process was carried out on the pre-processed dataset and the quality of the model was estimated by traditional metrics for regression models.

   Evaluation Metrics:

   - R^2 Score: 0.9744
   - MAE: 311,706,012.47
   - RMSE: 454,072,561.56

   Analysis:

   - The coefficient of determination of 0.9744 means that about 97.44% of variance is explained by the model, meaning high predictive power.
   - Despite being quite large, both MAE and RMSE are small compared to the scale of construction costs, making the model practically reliable.

- It can be concluded from the analysis that the model gives precise and reliable predictions.

2. **Actual vs Predicted Cost Analysis**

The correlation between actual cost and predicted cost was estimated with the help

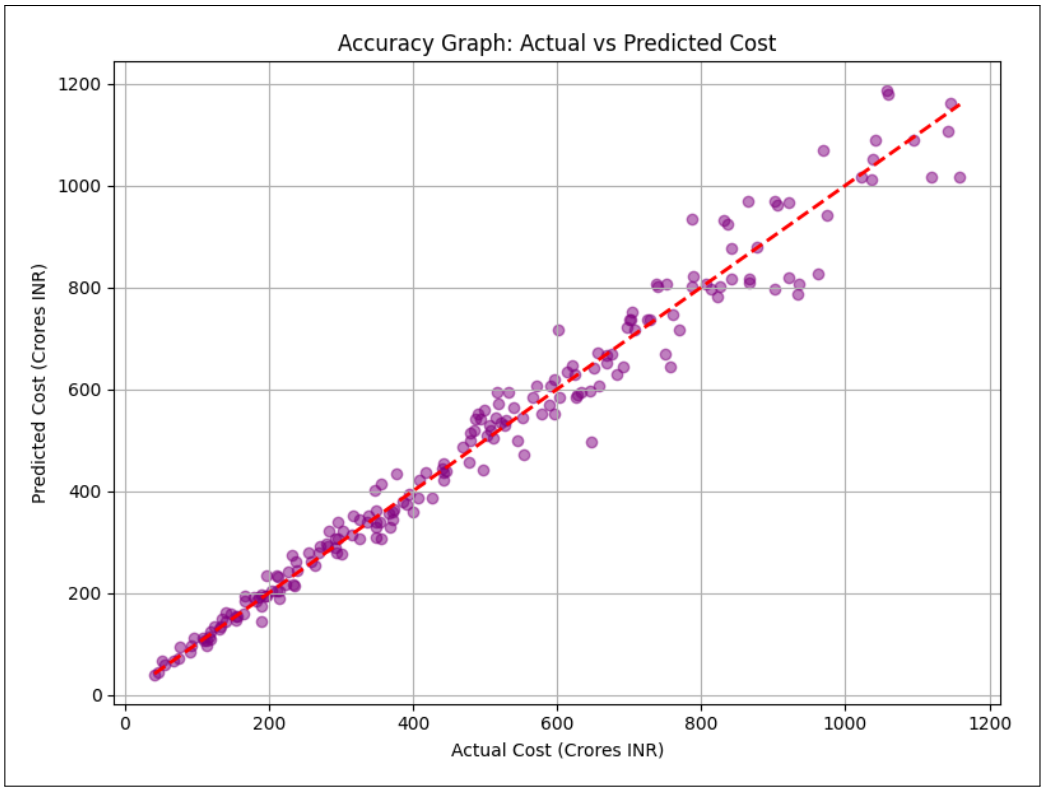


Figure 8.1: Accuracy graph

of scatter plots (Fig. 8.1.

- The correlation between actual and predicted values turned out to be quite high.
- The vast majority of points are located in proximity to the optimal prediction line, confirming the high precision of the model.
- The model demonstrates high consistency in predicting costs regardless of the scale of construction.
- Some differences may be noted when predicting the cost of very large constructions.

This indicates high reliability of the model in real-life situations.

3. **Residual Analysis**

Residual analysis (fig. 8.2)was applied to examine errors in predictions.

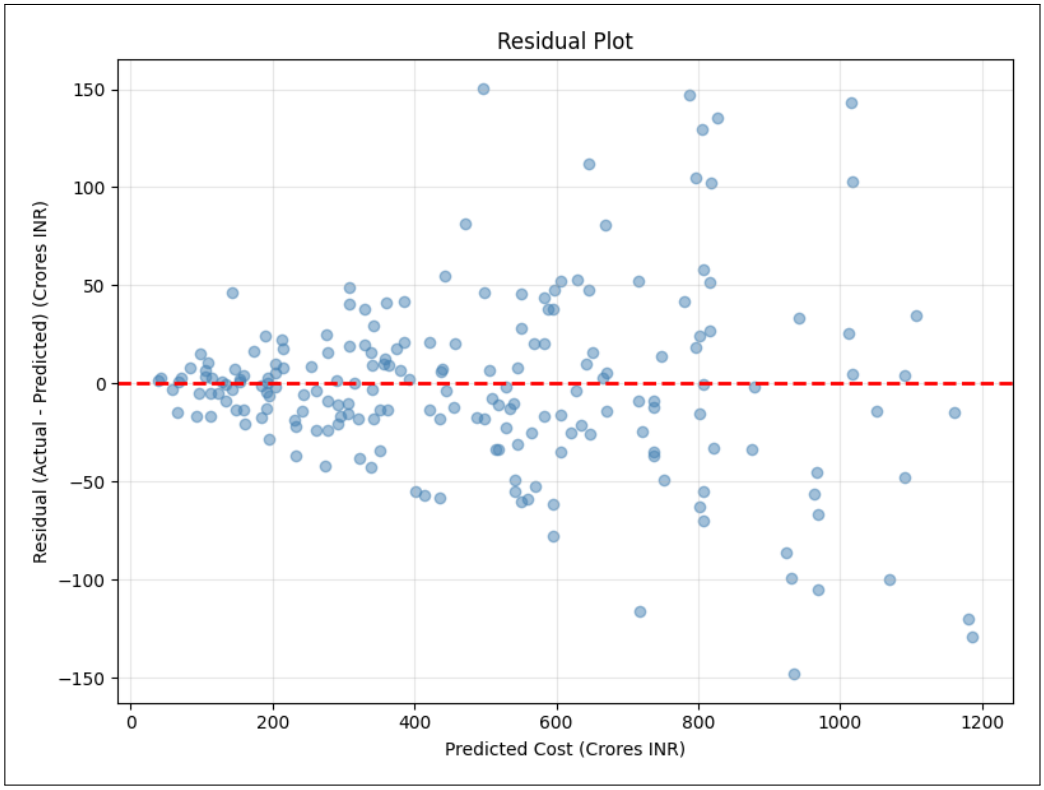


Figure 8.2: Residual Plot

- Random distribution of residuals around 0 proves absence of systematic errors.
- Pattern in prediction errors is not evident, which means that model underfitting and overfitting are absent.
- A few prediction errors occurred for high-cost projects, which is normal for any regression analysis on big datasets.

It proves that the developed model is accurate, consistent, and generalized.

4. **Feature Importance Analysis**

The feature importance analysis (Fig. 8.3 was carried out to understand which parameters determine construction costs.

- Features of primary importance are:
  - Project duration
  - Length of drainage pipes
  - Type and length of road
  - Number of bridges
- Secondary features are:
  - Labor resources needed

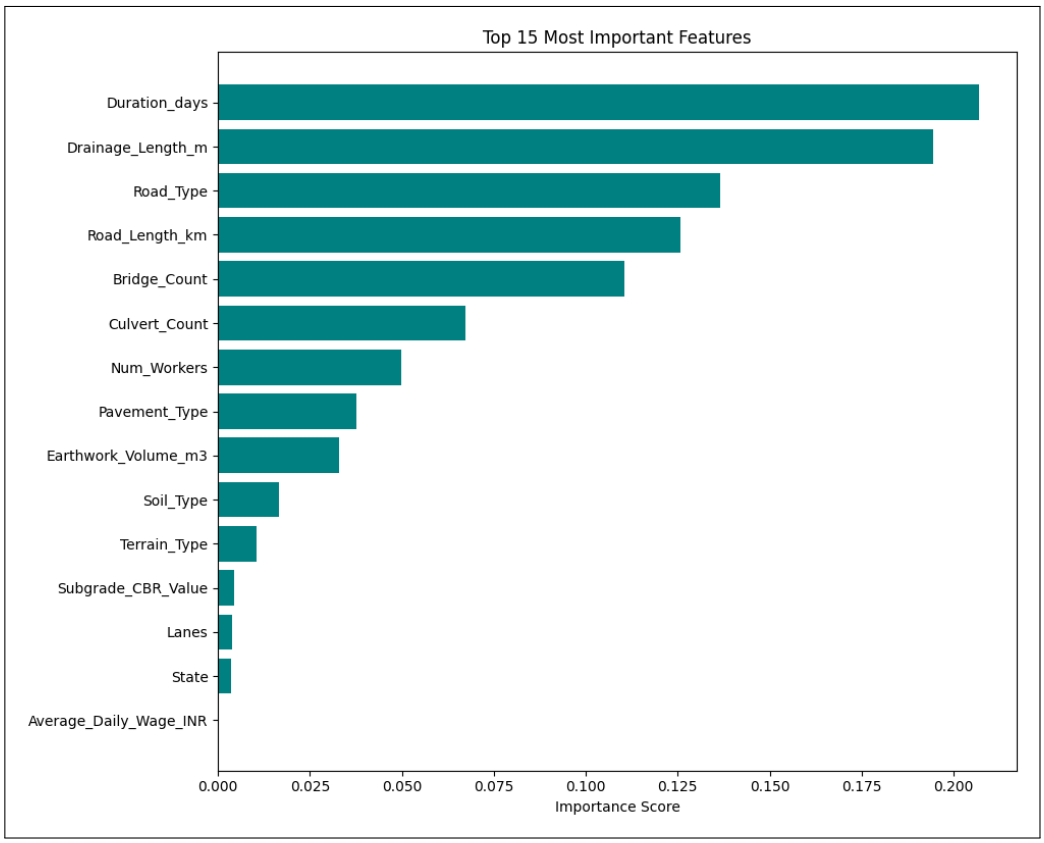


Figure 8.3: Important Features

- Pavement conditions
- Volume of earthworks

This analysis shows which factors affect costs significantly.

5. **Blockchain and Smart Contract Validation**

   - All transactions were stored immutably in blockchain storage.
   - Smart contracts ensured proper milestones-based funds release.
   - Funds could not be withdrawn by other participants because of smart contract logic.
   - Details of transactions were stored and successfully retrieved (hash, time stamp, transaction amount).

   Thus, the blockchain module proves its functionality.

6. **System Integration Analysis**

   - Frontend, backend, blockchain, and machine learning modules were successfully integrated.

- REST API and Web3 ensured data interchange between modules.
- No problems related to data inconsistency were observed.

Hence, the platform is stable and integrated.

7. **System Performance Evaluation**

   - System responses were timely when performing tasks like login, creating projects, and updating information.
   - Blockchain transaction processing was within an acceptable timeframe.
   - The system performed various tasks simultaneously without any loss in performance.

   This is proof that the system satisfies performance and usability criteria.

Validation:

- All features were validated by implementing the specified test cases.
- All output values conformed to the required results in each module.
- The application was evaluated under different conditions to check its reliability.
- There were no serious errors during evaluation.

Conclusion:
The obtained findings show that the InfraSync framework works efficiently within all subsystems. The XGBoost model showed outstanding results (R^2 = 0.9744), indicating its ability to predict costs with high accuracy. The blockchain framework guaranteed the transparency and proper management of funds, and the flawless integration provided efficient operations. Thus, the use of both machine learning and blockchain technologies can be considered effective in improving the transparency, efficiency, and decision-making processes of road construction management.

8.3 Screenshots

Below are some examples that show the functioning of the InfraSync infrastructure through screenshots and output:

1. System Homepage.



Figure 8.4: Homepage

- Provides the system homepage interface. (fig. 8.4
- Provides user roles selection including Government Official, Contractor, and Public Transparency.
- Serves as an interface to access the modules in the system.

2. Government Portal - Tender Creation

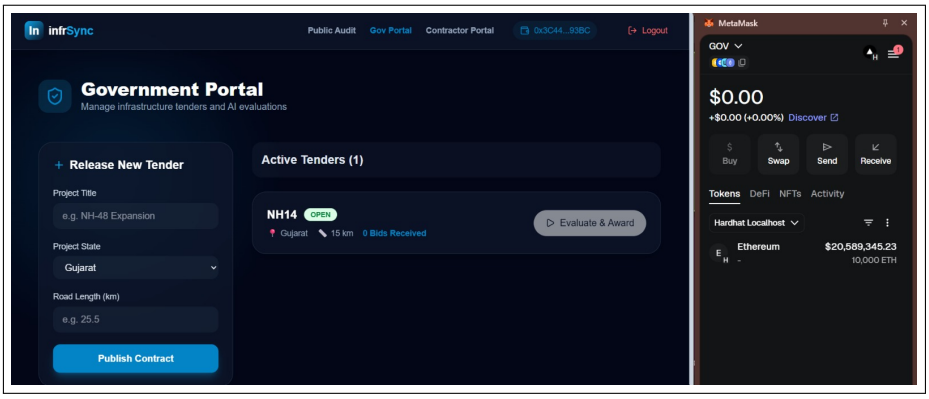


Figure 8.5: Tender Creation

- Provides the interface for creating new tenders.(Fig. 8.5
- Provides the opportunity for the administrator to input projects' details like project title, state, and length of the road.
- Starts the process of tender creation and showing active tenders.

3. Bid Evaluation Interface

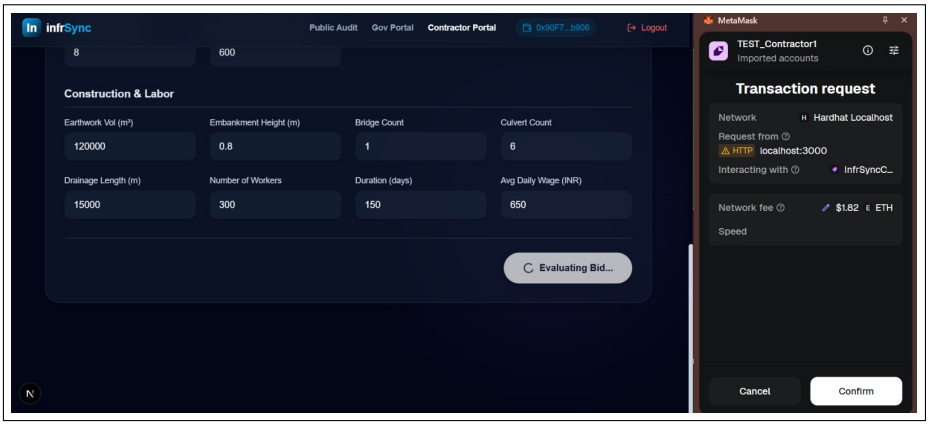


Figure 8.6: Bid Evaluation Interface

- Fig. 8.6 Shows evaluation of bids from contractors using the machine learning model.
- Comparing the bid amount with the predicted cost produced by the machine learning model.
- Aids in selecting the best contractor based on the results of the analysis.

4. Contract Awarding Screen

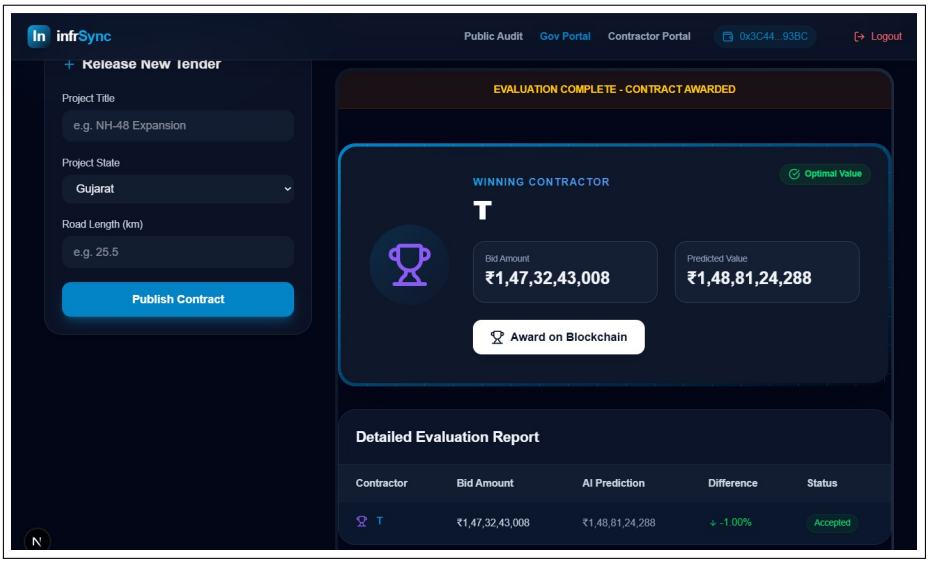


Figure 8.7: Contract Award

- Fig. 8.7 Provides the final stage where the contract is awarded.
- Shows details of the selected contractor like bid amount and the predicted cost.
- Secures awarding of the contract using blockchain technology.

5. Public Transparency Portal

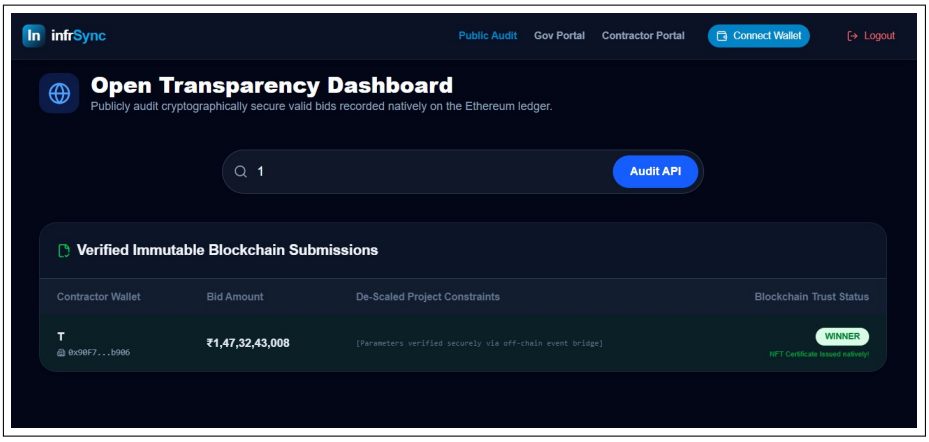


Figure 8.8: Public Transparency Portal

- Fig. A.1 Provides the transactions data which can be accessed publicly.
- Shows the contractor wallet, bid amount, and whether it was verified.
- Ensures transparency in awarding the contract using blockchain.

6. Contractor Reputation (NFT Certificate)

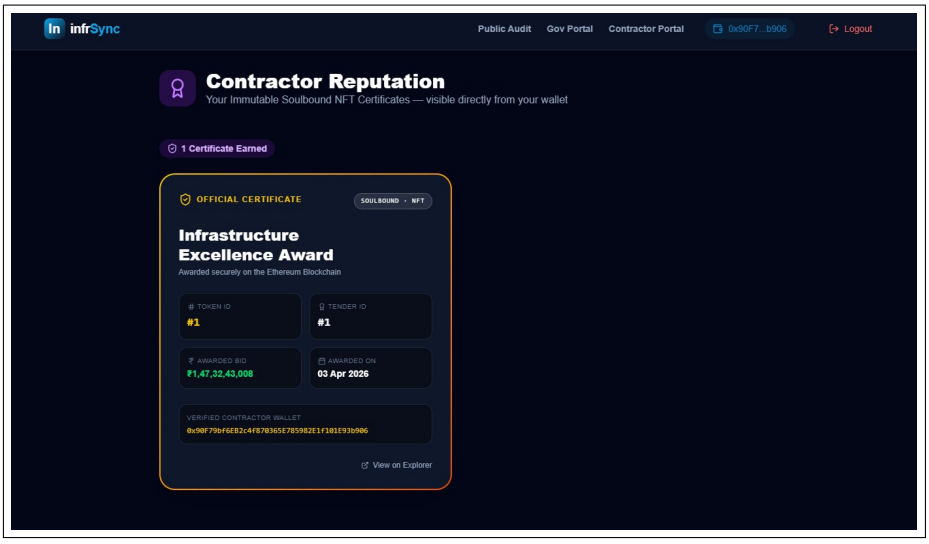


Figure 8.9: NFT Certificate

- Fig. 8.9 Provides NFT-based certificate awarded to the contractor.
- Shows details of the NFT such as the token ID, tender ID, amount awarded, and date.

7. Blockchain Transactions (MetaMask)

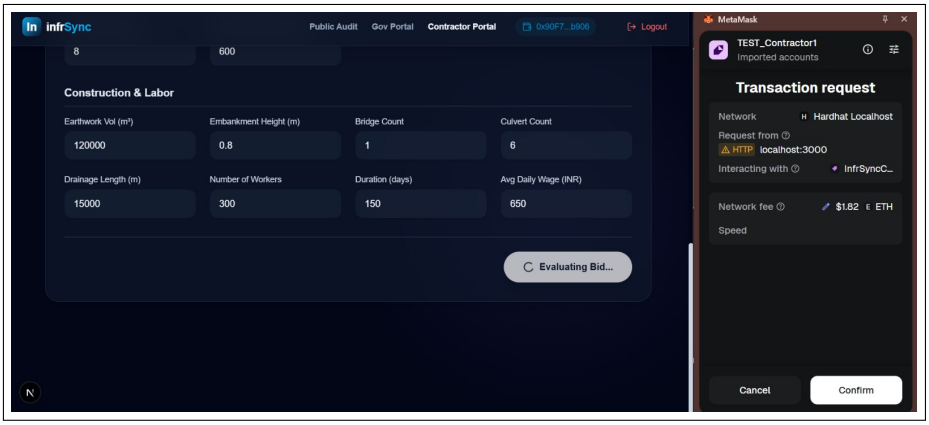


Figure 8.10: Blockchain Transaction

- Fig. 8.10 Shows transaction confirmation through the MetaMask wallet
- Illustrates information about the network, transaction fees, and smart contract actions.
- Guarantees secure transaction of transactions on the blockchain.

Chapter 9

CONTRIBUTION TO SUSTAINABLE DEVELOPMENT GOALS (SDGS)

9.1 Introduction to SDGs

Sustainable Development Goals (SDGs) are global goals set forth by the United Nations that focus on solving pressing issues such as infrastructure development, economic growth, and sustainable urbanization. All 17 SDGs have a single purpose – to guarantee that everyone has an equal opportunity to achieve sustainable development by 2030.
The two most important of those goals are infrastructure development and sustainable urbanization since they have a positive impact on improving people's quality of life and helping businesses thrive. The InfraSync system can help with both goals due to the usage of cutting-edge blockchain and machine learning technologies.

9.2 Mapping of the Project to Relevant SDGs

InfraSync system is mainly aimed at the following Sustainable Development Goals:

SDG 9: Industry, Innovation, and Infrastructure

This goal benefits greatly from the use of InfraSync system because of multiple reasons listed below.

- First of all, infrastructure projects will be conducted efficiently and reliably because of the appropriate use of funds in accordance with the contract.
- Secondly, the usage of blockchain technology adds innovative features to a conventional approach to project management.
- Thirdly, smart contracts automate the financial process and minimize dependency on humans.
- Finally, machine learning allows to predict costs more accurately.

Using these technologies together, InfraSync facilitates the development of current, resilient, and sustainable infrastructure systems.

SDG 11: Sustainable Cities and Communities

InfraSync can contribute to SDG 11 by upgrading urban infrastructure and fostering sustainable urban development.

- Road constructions can be done more effectively, ensuring efficient transport networks.
- Projects will be completed on time, minimizing disruption to citizens.
- TEffective project management results in improved infrastructure that benefits society.
- Resource utilization becomes efficient, making urban infrastructure sustainable.

It is an important tool in the construction of sustainable cities by optimizing infrastructure development.

9.3 Evaluation of the Impact of the Project

InfraSync has various impacts on society which include:

Impact on economics:

- Saves money from financial wastage due to appropriate allocation of resources.
- Avoids losses from corrupt practices and improper management of funds.
- Enables more accurate cost predictions using machine learning algorithms.

Impact on Society:

- Increases public trust in government activities due to transparency.
- Improves road networks, hence benefiting people in their day-to-day activities.
- Promotes responsible behavior by all stakeholders during construction projects.

Impact on Technology:

- Shows practical application of blockchain technology in infrastructural projects.
- Advocates the use of new technological innovations in old systems.

9.4 Issues and Future Scope

Issues:

- The use of blockchain technology in a major governmental system could entail modifications to current policies.
- The integration process with pre-existing infrastructure systems may present difficulties.
- Data accuracy for machine learning predictions is significant.
- System scalability should be considered.

Future Scope:

- Real-time connection to the governmental system database.
- Implementation in other areas of infrastructure like bridges and smart cities.
- Improving the prediction model with an extensive database.
- Scaling up the system with enhanced system efficiency.
- Mobile-based system accessibility.

Chapter 10

CONCLUSION & FUTURE SCOPE

10.1 Conclusion

InfraSync system successfully solves multiple problems associated with road construction management, which include poor transparency, inefficient resource allocation, and imprecise cost calculation. Thanks to blockchain technology, all financial transactions related to construction will be clear, secure, and reliable both for the developers and the investors. Due to implementation of smart contracts, automatic payment of the money is ensured through a milestone-based approach whereby the release of the funds occurs after the accomplishment of certain milestones. This process is automatically carried out, meaning that no human interaction is required during payment procedures. In addition, thanks to XGBoost machine learning algorithm that is implemented into the system, cost prediction becomes highly precise since all the factors, including cost of materials, labor efforts, and magnitude of the project, are taken into account. Finally, an easy-to-use interface is developed through creating the frontend layer, making monitoring and analyzing transactions possible for any stakeholder without having sufficient technical knowledge. Integration of different types of technologies within one system proves that there is potential in creating systems where multiple technologies are involved, and their interaction results in the solution of specific issues. To conclude, InfraSync is highly effective in fulfilling its goals.

10.2 Future Work

Although InfraSync has been effectively designed as a prototype, there is scope for improvement and development of the software to make it more reliable, scalable, and deployable in reality. These modifications would help expand the potential uses of the system on a practical basis.

1. **Connection with Government Databases**
   Connection can be made between the current InfraSync and government databases or infrastructure management systems for real-time tracking of the progress and performance of infrastructure projects.

- Real-time connection would facilitate higher accuracy through use of actual data from governmental agencies.
- Would allow for central monitoring of different projects and initiatives undertaken in various regions.
- Helps in administration-level decision-making and governance.

2. **Improvements in ML models** The reliability and effectiveness of cost prediction can be improved by refining the machine learning models used by InfraSync.

   - Larger and more diverse training dataset should be used when training the model.
   - Inclusion of additional relevant features such as environmental factors, availability of materials, and previous data from other similar projects.
   - Hyperparameter tuning and other methods can be used for optimization of the model.

3. **Blockchain Scalability:**
   For widespread implementation, the blockchain technology should be modified to handle increased traffic more efficiently.

   - Using second-layer technologies or sidechains to minimize transaction processing time.
   - Optimization of the smart contracts' code for reduced gas fees.
   - Performance management of the system under load conditions.

   It will allow nationwide or wide-spread implementation.

4. **Improvements to User Experience**
   The user interface could be redesigned to improve the user experience.

   - Dashboard redesign with visual analytics and graphs
   - Navigation improvements and faster response time
   - Optimized workflow for non-expert users

   Improving user experience will increase system usability.

5. **Development of the Mobile App**
   A mobile application could be created to increase accessibility.

   - Monitoring projects and making transactions on the move

- Notifications about milestones achieved and payments made
- Convenience of use for field users and authorities

6. **Analytics and reporting enhancements** Further analytical capabilities could be implemented into the system.
   - Reporting on the results and progress of different projects
   - Predictive analytics to forecast delays and other risks
   - Project comparison

   This will help to increase decision support capabilities.

7. **Additional Security Improvements**
   Some additional security measures that may be implemented to improve the security level even more include:
   - Using advanced systems of user authentication, like multi-factor authentication
   - Implementing more secure encryption algorithms to protect sensitive data
   - Continual monitoring of the system to detect possible threats to security

   These steps will allow ensuring the highest levels of reliability and security of the system.

10.3 Applications

The InfraSync system can find a wide variety of application areas in practice since the implementation of blockchain and machine learning in the infrastructure sphere opens many opportunities for its adaptation for solving practical tasks in various industries. Below, some potential application spheres are considered:

1. **Public Projects on Construction of Roads, Highways, Bridges, Utilities**
   The InfraSync system can be applied in all government-related infrastructure projects in order to provide more transparent spending and management of public money.
   - Provides more transparent management of funds
   - Facilitates real-time project monitoring
   - Helps reduce corruption in infrastructure projects
   - Aids efficient implementation of infrastructure projects

2. **Building of Smart Cities**
   The InfraSync system can be also used for implementing infrastructure-related projects in smart cities.

   - Allows organizing smart urban development projects efficiently
   - Promotes better coordination among departments in charge of infrastructure projects
   - Guarantees higher quality of infrastructure projects

3. **Construction Sector** The InfraSync solution can be implemented in privately-owned construction firms to optimize project management and financial oversight.

   - Facilitates project tracking and milestones achievement
   - Guarantees proper budget utilization
   - Eliminates delays and optimizes project management
   - Promotes coordination among project participants

4. **Financial Tracking Systems**
   he blockchain-based technology utilized in InfraSync can be employed in any financial tracking system requiring secure and verifiable transaction tracking.

   - Permits immutable financial transactions logging
   - Provides audit trails
   - Prevents financial fraud and mismanagement

   Can be employed in industries such as banking, insurance, and government accounting

5. **Infrastructure Auditing Systems**
   The InfraSync system can support auditors to verify project implementation and budget management.

   - Offers immutable financial transactions logging
   - Facilitates auditing via blockchain information
   - Detects anomalies in financial transactions
   - Assists with regulatory compliance

The flexibility of the InfraSync application means that it could have several other uses beyond those related to the construction of roads. Transparency, automation, and

accuracy make the solution relevant for current project management solutions used in infrastructure. If improved even more, the technology could become a full-fledged platform that would help manage large and complicated projects in various industries.

Appendix A

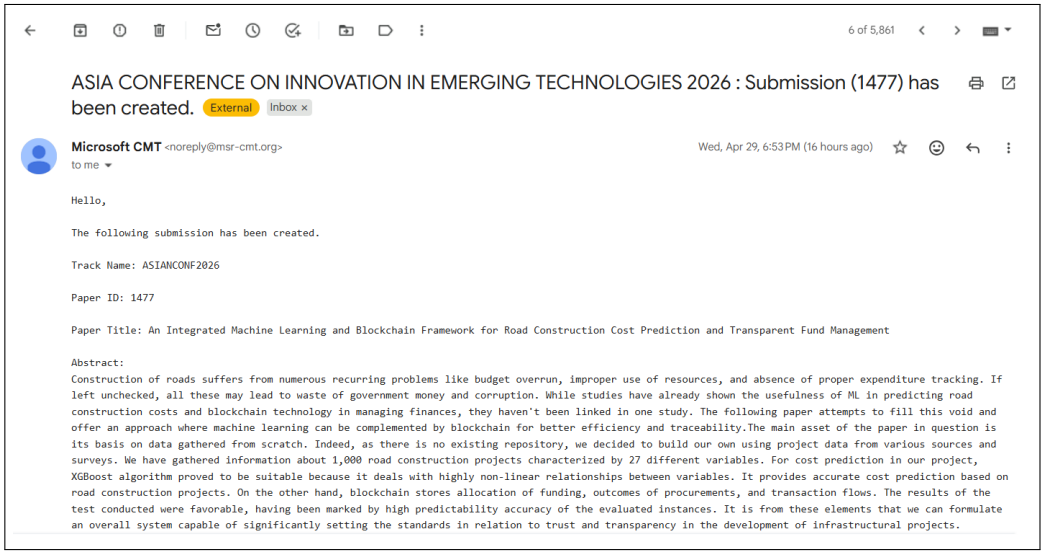
Details of The Research Paper

A.1 Paper Details

Title:	An Integrated Machine Learning and Blockchain Framework for Road Construction Cost Prediction and Transparent Fund Management
Authors:	Om Sagar, Parag Pawar, Harshal Suroshe, Vinayak Suryawanshi, Rucha Shinde
Conference:	Asia Conference on Innovation in Emerging Technologies 2026
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Paper Title: An Integrated Machine Learning and Blockchain Framework for Road Construction Cost Prediction and Transparent Fund Management

Abstract:
Construction of roads suffers from numerous recurring problems like budget overrun, improper use of resources, and absence of proper expenditure tracking. If left unchecked, all these may lead to waste of government money and corruption. While studies have already shown the usefulness of ML in predicting road construction costs and blockchain technology in managing finances, they haven't been linked in one study. The following paper attempts to fill this void and offer an approach where machine learning can be complemented by blockchain for better efficiency and traceability.The main asset of the paper in question is its basis on data gathered from scratch. Indeed, as there is no existing repository, we decided to build our own using project data from various sources and surveys. We have gathered information about 1,000 road construction projects characterized by 27 different variables. For cost prediction in our project, XGBoost algorithm proved to be suitable because it deals with highly non-linear relationships between variables. It provides accurate cost prediction based on road construction projects. On the other hand, blockchain stores allocation of funding, outcomes of procurements, and transaction flows. The results of the test conducted were favorable, having been marked by high predictability accuracy of the evaluated instances. It is from these elements that we can formulate an overall system capable of significantly setting the standards in relation to trust and transparency in the development of infrastructural projects.

Figure A.1: Email Confirmation of Paper Submission

Appendix B

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A Project REPORT ON

Infrasync : Blockchain and Machine Learning Based System for Transparent Fund Allocation and Road Construction Monitoring

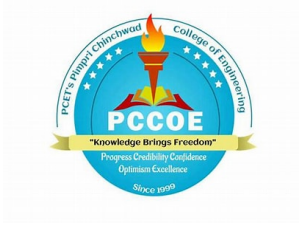
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FOR THE AWARD OF THE DEGREE

BACHELOR OF TECHNOLOGY
(COMPUTER ENGINEERING (REGIONAL LANGUAGE))

SUBMITTED BY

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DEPARTMENT OF COMPUTER ENGINEERING
(REGIONAL LANGUAGE)



CERTIFICATE

This is to certify that the project report entitled

INFRASYNC: BLOCKCHAIN AND MACHINE LEARNING BASED SYSTEM FOR TRANSPARENT FUND ALLOCATION AND ROAD CONSTRUCTION MONITORING

Submitted by

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VINAYAK SURYAWANSHI	**PRN : 122B1D062**

are bonafide students of this institute and the work has been carried out by them under the supervision of Prof. Rucha Shinde and it is approved for the partial fulfilment of the requirement of Pimpri Chinchwad College of Engineering, an autonomous institute, for the award of the B. Tech. degree in Computer Engineering.

(Prof. Rucha Shinde)
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(Regional Language)

(Prof. Dr. Rachana Y. Patil)
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Pimpri Chinchwad College of Engineering Pune – 44

Place : Pune
Date :

This Project report has been examined by us as per the requirements at Department of

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